NOTE: This document has been changed to reflect current practices. There are no other changes to this document. In addition, all documents associated with this document are hereby designated as standard practices.

INCH-POUND

MIL-STD-1325A(NAVY)
2 MARCH 1976
SUPERSEDING
MIL-STD-1325(NAVY)
13 MARCH 1972

DEPARTMENT OF DEFENSE
STANDARD PRACTICE

RAILCAR LOADING OF HAZARDOUS MATERIALS

AMSC N/A

FSC 8140

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DEPARTMENT OF THE NAVY
Naval Sea Systems Command
Washington, D C 20360

Railcar Loading of Hazardous Materials

MIL-STD-13[illegible]5A (Navy)

1 This standard has been approved by the Department of the Navy and is published to establish requirements for railcar loading of hazardous materials

2 As of the promulgation date of this document, this standard is a mandatory requirement to be invoked in all applicable specifications, purchase descriptions or military interdepartmental procurement requests (and contracts, when necessary) in the procurement of naval ammunition, explosives, and associated inert items to be transported by railcar It is mandatory for performance of railcar loading operations by all elements of the Navy and the Marine Corps

3 Recommended corrections, additions, or deletions should be addressed to the Commanding Officer, Naval Ordnance Station, Standardization Division Indian Head, Md 20640

4 Requests for technical interpretations or approval of deviations should be addressed to Commanding Officer, Naval Weapons Station Earle, Naval Weapons Handling Laboratory Colts Neck, N J 07722

5 Copies of this complete standard or individual dash sheets may be obtained from the Commanding Officer, Naval Publications and Forms Center 5801 Tabor Avenue, Philadelphia, Pa 19120

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FOREWORD

This standard describes general procedures and practices applicable to loading and securing hazardous materials for transportation by or to the Navy in railcars.

Different loads require different shoring applications. It is the intent of this standard not only to describe the special applications but also to set up minimum standards for the shoring of railcar loads. These procedures will help loading personnel prepare safe and economical loads.

Physical dimensions, weights, types of loads and vehicles vary greatly, precluding the coverage of all combinations. The examples and procedures given in this basic standard shall be considered as typical. Mandatory requirements for specific loads are given in a series of MIL-STD dash number sheets which also form a part of this standard.

The Association of American Railroads, through its Bureau of Explosives reviews and approves all carloading specifications for hazardous materials as assigned by the Department of Transportation and the Canadian Transport Commission.

A motion picture pertinent to this standard, entitled "Railroad Carloading and Bracing of Ammunition", MA-10715, is available for training purposes from your local Naval Education and Training Support Center/Detachment by submitting requests for Training Aids Temporary Loan Request/Invoice (5NC GEN 1551/1 Rev 9-74).

Certain related weapon requirements (WR's) referenced within this standard are in process of supersedure by military standards. If an equivalent military standard dash number sheet has not been published, the applicable WR slash number sheet shall be used. The following cross-reference is provided to facilitate the identification between the newly assigned designator and the previous designator.

New	Previous
MIL-STD-1320	WR-51
MIL-STD-1322	WR-53
MIL-STD-1323	WR-54
MIL-STD-1324	WR-55

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MIL-STD-1325C

MILITARY STANDARD

RAILCAR LOADING OF HAZARDOUS MATERIALS

1 SCOPE

1.1 This standard contains general and detail requirements for the preparation of full and less than carload shipments of ammunition, explosives, propellants, and weapon system components. It also contains guidance to be followed in all carloading procedures when specific instructions in the form of MIL-STD dash number sheets do not exist (for example, mixed carloads). The MIL-STD dash number sheets contain specific instructions primarily for carloading ammunition normally shipped in large quantities.

1.2 **Application.** This standard is limited to the preparation for shipment of material by or to the Navy. It applies to shipments from one Government activity to another and (where referenced in the applicable contract) shipment of material to the Navy from contractors' plants. It does not apply to interplant shipments of material that is not Government owned. MIL-STD-1325-100 and MIL-STD-1325-101 are typical specifications applicable to carloading palletized unit loads of many different components which do not require the detail shown by specific carloading plans (dash sheets). MIL-STD-1325-102 is a typical specification showing procedures authorized for retaining partial upper layers in all metal boxcars.

1.3 **Compliance with law.** The Code of Federal Regulations, Title 49, administered by the Department of Transportation, imposes clear responsibilities on shippers of hazardous materials including military shippers. It provides severe penalties against those who violate the regulations issued under this law. Compliance with the procedures outlined herein assures the individual shipper of compliance with his legal obligations.

1.3.1 In order to ensure compliance with law and regulations, this standard and its MIL-STD dash number sheets have been coordinated with or approved by the Bureau of Explosives.

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1.4 **Safety precautions** The safety precautions outlined in OP 3347 shall be observed. The following are general safety precautions that are not related to any specific procedure and therefore do not appear elsewhere in this standard. These precautions are recommended safety precautions that personnel shall understand and apply during many phases of carloading.

1.4.1 **Materials handling equipment** Never exceed the rated allowable load of any piece of materials handling equipment.

1.4.2 **Gangways, bridge plates, etc.** Always be sure gangways, bridge plates and other means of entering cars are of sufficient strength to carry the load safely. Consider the gross load such as the forklift truck plus the load.

1.4.3 **Ammunition and explosive loading** Safety and responsibility are the paramount considerations in loading ammunition and explosives. Know the materials you are handling and remember carelessness can and will kill YOU.

2. REFERENCED DOCUMENTS

2.1 The issue of the following documents in effect on the date of invitation for bids form a part of this standard to the extent specified herein.

SPECIFICATIONS

Federal

FF-N-105	Nails, Brads, Staples, and Spikes: Wire, Cut and Wrought
MM-L-736	Lumber, Hardwood
MM-L-751	Lumber, Softwood
QQ-S-781	Strapping, Steel, Flat and Slats
PPP-F-320	Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes

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2. APPLICABLE DOCUMENTS

Military Handbook

MIL-HDBK 236	Index to Standards for Palletizing Truck Loading, Railcar Loading, and Container Loading of Hazardous Materials

Naval Sea Systems Command (Code Ident 10001)

OP 5	Ammunition and Explosives Ashore
OP 1750	Freight Car Mechanical Dunnage Systems Pin and Rack Type, Instructions for Operation and Use
OP 2165	Navy Transportation Safety Handbook
OP 2173	Handling Equipment for Weapons and Explosives Mobile Equipment
OP 3347	United States Navy Ordnance Safety Precautions
OP 3681	Motor Vehicle and Railcar Load Inspector's Manual for Ammunition, Explosives and Other Dangerous Articles
OP 4098	Handling Ammunition and Hazardous Material With Industrial Trucks
NAVORD FORM 8023	Railroad Car Inspection Report

Naval Supply Systems Command

NAVSUP 444	Military Traffic Management Regulations

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** The following documents form a part of this standard to the the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids shall apply.

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Department of Agriculture

Agriculture Handbook No 72	Wood Handbook

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D C 20402)

Department of Transportation

Code of Federal Regulations

49 CFR 100-199	Transportation

(Applications for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D C 20402 Orders for the publication should cite "the latest issue and supplements thereto ")

R M Graziano's Tariff No 30	Hazardous Materials Regulations of the Department of Transportation

(Applications for copies should be addressed to R M Graziano Agent, 1920 L St N W Washington, D C 20036)

Association of American Railroads (AAR)

*Pamphlet No 6	Methods for Loading and Bracing Carload and Less than Carload Shipments of Explosives and Other Hazardous Materials
*Pamphlet No 6A	Illustrating Methods for Loading and Bracing Carload Shipments of Military Ammunition and Explosives
**Pamphlet No 14	Rules Regulating the Safe Loading of Freight in Closed Cars and Protection of Equipment
**Pamphlet No 27	Rules Regulating the Preparation and Safe Loading of Empty Projectiles, Bombs and Cartridge Cases in Closed Cars and Protection of Equipment
**Circular No 42F	General Rules Covering Loading of Carload Shipment of Commodity in Closed Cars

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** Manual of Standards and Recommended Practices, Section [illegible] Procedures Governing the Loading of Commodity or Open Top Cars

(Applications for copies of AAR Documents should be addressed as follows *Association American Railroads, Bureau of Explosives, 1920 L St., N.W. Washington, D.C. 20036 **Association of American Railroads, Freight Loading and Container Section, 59 E. Van Buren St., Chicago, Ill. 60605 ***Association of American Railroads, Mechanical Division 59 E. Van Buren St., Chicago, Ill. 60605.)

3. DEFINITIONS

3.1 **General.** The definitions given herein cover terms as they are used in this standard and are not to be confused with any definitions appearing elsewhere.

3.2 **Anchor plate.** A steel plate used to anchor steel strapping to a railcar. It is slotted to receive the strapping and drilled to permit nailing to railcar walls or floor.

3.3 **Bay.** One or more stacks in a boxcar separated by the remainder of the lading by crossmembers or by divisional gates.

3.4 **Block.** A bulky, usually solid piece of wood with one or more flat faces.

3.5 **Block, chock.** A concave or beveled block of wood used to secure lading in position. See figure 1.

3.6 **Brace.** A structural member used to transmit, divert, or resist weight or pressure. In carloading, the word is usually modified by a functional description.

3.7 **Brace, cross.** A single member, wood or wood and metal combined, placed crosswise in a car against the lading to secure the lading in position.

3.8 **Brace, sway.** A piece or assembly used to prevent sideways motion of the lading resulting from lateral sway of the car.

3.9 **Bridge plate.** A bridge layed between a railcar and a loading dock or between cars to facilitate access to the car for loading or unloading the lading.

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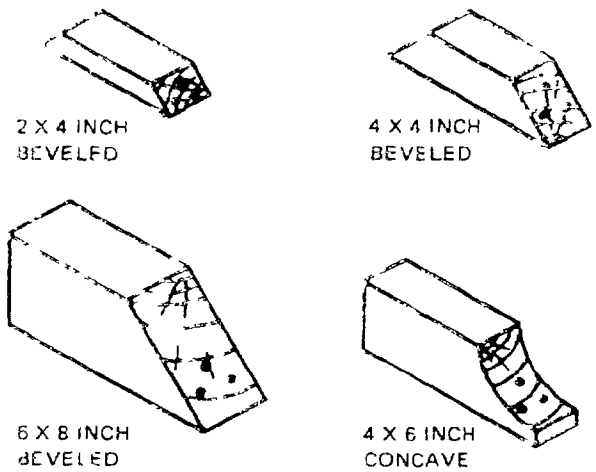


FIGURE 1 CHOCK BLOCKS

3 10 **Buffer strip or board** A piece of lumber placed against a container or piece of lading to provide a wide bearing surface to protect a sharp edged or thin walled item during transit

3 11 **Bureau of Explosives** The regulatory body of the Association of American Railroads responsible for the issuance and approval of appropriate rules for safety for shipment of explosives and hazardous materials by the railroad

3 12 **Capacity** The nominal capacity of a railcar expressed in round numbers for example 100 000 pounds (not to be confused with load limit)

3 13 **Car** A vehicle suitable for the carriage of freight by railroad Cars used for carriage of hazardous materials by rail may be of several types as follows

3 13 1 **Boxcar** A fully enclosed railroad car having a door or doors on both sides and, sometimes, on one or both ends Used for general freight service

3 13 2 **Commercial boxcar** A boxcar owned by one of the nation s railroads

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3.13.3 DODX boxcar A boxcar owned by the Department of Defense

3.13.4 **Class A car** A car which has been inspected and certified for carrying of Class A explosives in accordance with section 174.525 of the Code of Federal Regulations 49 CFR 100-199

3.13.5 **Class B car** A car suitable for the carriage of Class B explosives as defined in section 174.529 of the Code of Federal Regulations 49 CFR 100-199

3.13.6 **Class C car** A conventional boxcar suitable for the carriage of Class C explosives in accordance with the requirements of section 174.530 of the Code of Federal Regulations 49 CFR 100-199

3.13.7 **DF-type car** A special equipped boxcar known as dunnage free, having cross members as permanent load securing devices which are attached to steel rails attached to the sidewalls of the car. DODX boxcars, Series 29000, are of the DF type. Some DF-type cars have special cushioned draft gear to reduce shocks transmitted to the load.

3.13.8 **Double-door car** A car with a pair of doors on each side.

3.13.9 **Double-walled car** A boxcar with both sheathing and lining.

3.13.10 **End-door car** A boxcar with doors in each end. Not to be used for ammunition or explosives.

3.13.11 **Flatcar** An open car without roof, side, or end walls.

3.13.12 **Gondola car** An open car without roof, with low side and end walls.

3.13.13 **Open car** A car without a roof.

3.13.14 **Plug-door car** A car equipped with doors that close flush with the inside walls of the car. On each side of a plug door boxcar may be equipped with a single plug door, double plug doors, or one plug door and one conventional door.

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3.14 **Car lining.** A surface, usually wood, fastened to the inside of the car structure.

3.15 **Carload plan.** A specific design of the physical arrangement of lading and dunnage to protect particular items of lading from the hazards of rail transport.

3.16 **Car sheathing.** May be the same as car lining or, in the case of a double sheathed car, the boxcar will have an inside car lining and an outside sheathing which may be either wood or steel.

3.17 **Cleat.** A member, or group of members, used to reinforce another member or to hold it in its position. Cleat is usually modified by a functional description.

3.18 **Cleat, backup.** A reinforcing piece nailed to car floor or walls to secure blocks or braces.

3.19 **Cleat, holddown.** A cleat nailed to car walls to prevent braces or gates from rising. See figure 2.

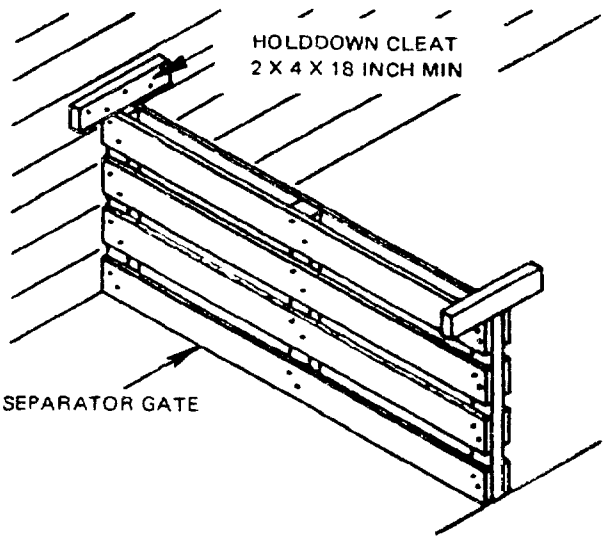


FIGURE 2. SEPARATOR GATE WITH HOLDDOWN CLEATS.

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3.20 Cleat, pocket [illegible] group of two or more cleats arranged to form a pocket to receive and restrain a cross brace or holddown member. See figure 3.

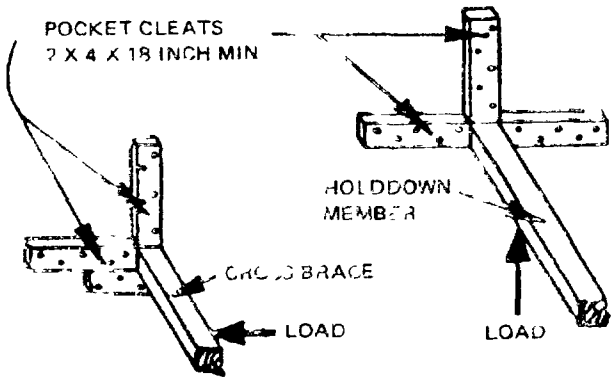


FIGURE 3. POCKET CLEATS

3.21 **Cleat, strut** A horizontal member oriented crosswise to the car and fastened to vertical gate members to serve as support for struts.

3.22 **Crossmember** A wood and steel member installed across the width of a DODX car, Series 29000, or DF equipped commercial cars to secure the lading in place.

3.23 **Crosspiece** A horizontal piece of wood in a center gate, end gate, or other space filler, extending across the width of the car. It may be placed directly against the lading or may hold or be held in position by the uprights that are against the lading. Also called a horizontal gate member.

3.24 **Doorway member** A steel and wooden member installed across the doorway of a DODX car, Series 29000, to permit installation of crossmembers across the car in the doorway area.

3.25 **Doorway protection** Dunnage material in or spanning the doorway to prevent the lading from falling or rolling out at doorway or coming in contact with side doors.

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3.26 Dunnage Any material (such as boards planks or blocks) used in transportation and storage to support and secure the lading to protect it from damage or for convenience in handling

3.27 Filler Material, usually boards, or frames used to fill space throughout the load in order to provide a smooth bearing surface or to compensate for irregularities in the lading or car

3.28 Gate A structure placed crosswise in the car and used to distribute the load or to fill space not occupied by lading Gates may be of the various types as indicated in 3.29 through 3.31

3.29 **Gate, center** A structure usually located in the doorway area of a boxcar separating the lading in each end of the car

3.30 **Gate, end** A structure placed against the end wall of a car to take up lengthwise space in a through load or, to distribute the load more evenly over end wall, or to protect the lading in cars with unlined corrugated steel ends

3.31 **Gate, separator (intermediate or divisional)** A structure used to facilitate transmittal of longitudinal forces from one stack to another and/or to separate the stacks of lading into sections throughout the car

3.32 **Hazardous materials** "Hazardous materials" means "explosives and other dangerous articles" as established by the Code of Federal Regulations 49 CFR 100-199

3.33 **Holddown member** A member secured by cleats used to prevent upward movement of the lading, gate, or bracing structure

3.34 **Lading** The load or cargo in the railroad car

3.35 **Layer** A course, or stratum of the load parallel to the floor of the car and one container or one unit load in height See figure 4

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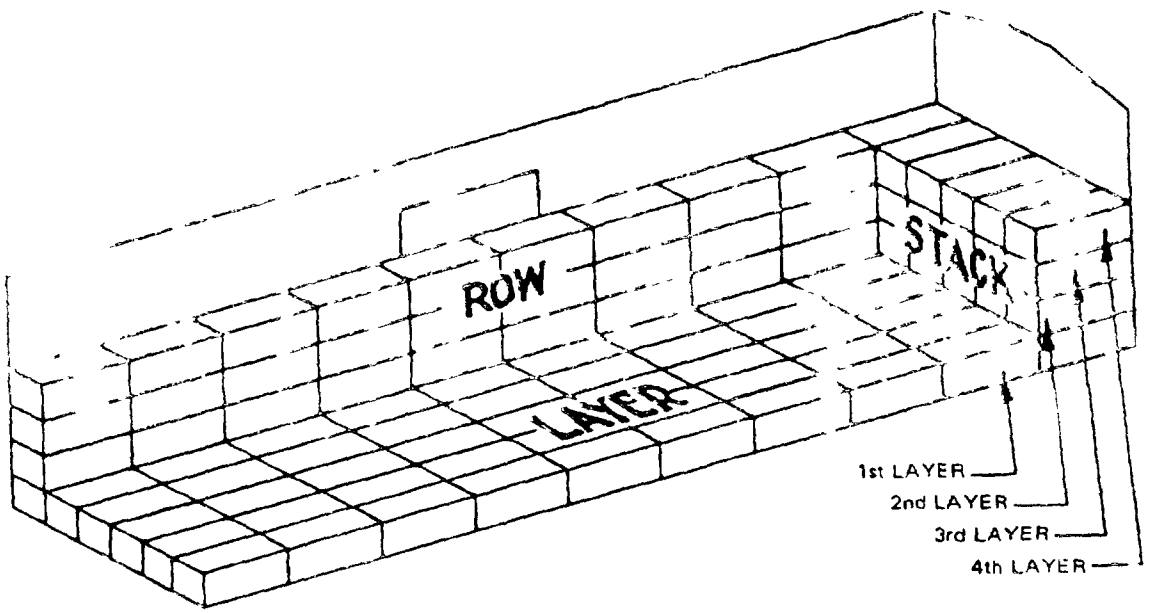


FIGURE 4 PARTIAL CARLOAD SHOWING NOMENCLATURE

3.36 **Load, divided** A load separated into two or more units by a center gate assembly or separator gates, or both

3.37 **Load limit** The greatest allowable weight which may be loaded into a railcar

3.38 **Load, palletized unit** Several (usually like) containers bound to a pallet to facilitate handling shipment, and storage

3.39 **Load, through** A load in which the lading extends through the doorway area not separated by center gates or divisional gates for the full length, or almost the full length of the car

3.40 **Nail finder** A light rake or board having a metal edge used to drag over the car lining in order to find protruding nails and staples that might damage lading

3.41 **Protector, stake pocket** A metal pad used in a stake pocket of a flat car to prevent the tiedown straps from wearing through

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3.42 **Row** A pile of containers or pallet loads extending lengthwise in the car parallel to the sides of the car and one container or unit load in width See figure 4

3.43 **Sleeper (floor blocking)** Member secured to the car floor, running in a lengthwise direction, and positioned against the lading to prevent cross movement of the load

3.44 **Stack** A pile of containers of unit loads, extending from one side of the car to the other, parallel to the end of the car and one container or unit load in length See figure 4

3.45 **Stiffeners** Wood members used to reinforce bracing

3.46 **Stringer** Members secured to the car floor or placed under or between layers of lading, running lengthwise of the car and used to support or to provide a supporting surface for a load

3.47 **Strut** A wooden member in a gate, lengthwise of the car, that spreads or separates the two faces of a gate

3.48 **Technical directing activity (TDA)** An activity designated by the cognizant systems command headquarters by contract, task assignment, or project order to assume responsibility for performing, directing, or monitoring the design and test of packaging, packing, shipping, handling, and transportation equipment for weapon system components

3.49 **Tie bars** Members used in center gates (with strut spans of 4 feet or more) to reduce tendency for failure of the strut by buckling in compression

3.50 **Unit load** A container or item or aggregate of either bound together to facilitate handling by mechanized equipment

3.51 **Wall member** A detachable member which fastens to the wall of a DODX car to locate and hold crossmembers

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4 GENERAL REQUIREMENTS

4.1 **General** This section covers the general requirements for the safe transportation of naval ammunition and hazardous materials by railcar. Detailed requirements are contained in section 5.

4.2 **MIL-STD dash number sheets** Specific instructions pertaining to the loading of specific ordnance items are contained on the MIL-STD dash number sheets. These sheets are identified by using a dash number following the basic MIL-STD-1325 designator. Sheets are individually numbered and titled. Where a MIL-STD dash number sheet exists, the loading, blocking, and bracing procedures shown in the dash number sheet shall be followed without exception for full carload shipments. The general rules on the selection of lumber for various functions and on nails and nailing given in this general standard shall be used as a guide. MIL-STD-1325-100 and MIL-STD-1325-101 are "Typical Carloads for Palletized Unit Loads." These sheets are applicable only to palletized unit loads. These sheets should be utilized when listed in the index for the particular item and may be used as a guide in carloading palletized unit load when no specific sheets exist. MIL-STD-1325-102 is a typical document showing dunnaging procedures for partial upper layers in all metal boxcars. When less than a full carload is to be shipped and a MIL-STD dash number sheet is available covering only a full carload, then the loading shall conform to the sheet except that the short quantity upper layers shall be secured in the car in accordance with the partial layer bracing procedures of 5.9.8.

4.2.1 **MIL-HDBK-236** MIL-HDBK-236 provides an index of items covered by the dash number sheets. This handbook titled "Index to Standards for Palletizing, Truckloading, Railcar Loading, and Container Loading of Hazardous Materials" provides an index to MIL-STD-1325 dash documents (carloading) in addition to the documents in the other areas listed in the title. The handbook includes three types of listings.

Section 1 lists, in alpha-numerical sequence, DODIC/NALC designated items that have "specific" or "typical" dash number sheets authorized for carloading of the items listed.

Section 2 lists, in alphabetical order, all the ammunition and weapon system component items that have "specific" or "typical" dash number sheets authorized for carloading of the items listed.

Section 3 lists numerically, all dash number documents giving the revision and change notice status of each document.

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NOTE

Users of MIL-STD-1325 dash number sheets shall consult Section 3 of the latest revision of MIL-HDBK-236 to confirm that they are using up-to-date dash number sheets

4.2.2 **Variations or deviations** Prior to variations or deviations to the basic procedures the details of the proposed changes shall be discussed with and approved by the Naval Weapons Station Earle, Naval Weapons Handling Laboratory (NWHL), Colts Neck, N J 07722 (phone autovon 440-7693)

NOTE

It is anticipated that the individual commands shipping ammunition and explosive materials will develop a cooperative working relationship with the representative of the Bureau of Explosives and NWHL in order that safe shipping practices will be universally observed

CAUTION

All carloading designs for naval ammunition and explosives are subject to the approval of TDA and NWHL

4.2.3 **Procedures to be followed when no MIL-STD sheet exists** When none of the dash sheets contained in this standard can be applied to the item to be shipped, including typical carload sheets specified in 4.2, or the hazard classification of the items to be shipped is not known the following subparagraphs shall apply in the order listed

4.2.3.1 **Determination of hazard classification** The hazard classification of the lading shall be determined prior to the promulgation of any carloading plan in accordance with the Department of Transportation (DOT) regulations 49 CFR 100-199 or OP 2165

NOTE

The DOT regulations are constantly being revised The R M Graziano Tariff contains the DOT regulations and is periodically republished with current changes and updated requirements Shipping activities are expected to subscribe to this Tariff to assure availability of the latest issue

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In addition to explosives flammable materials oxidizing materials corrosive liquids, poisons, and radioactive materials are also covered by the DOT regulations In determining whether or not an item is explosive or dangerous consult OP 2165, or the R M Graziano Tariff

4 2 3 2 In the event the product is not clearly definable as dangerous or nondangerous after reading these references, contact the Naval Sea Systems Command (Code SEA-04H7) for appropriate advice

WARNING

Do not ship explosives or dangerous materials unless the proper DOT hazard classification has been assigned

CAUTION

New explosives except samples for laboratory examination, may not be legally shipped unless the Bureau of Explosives or the Department of Defense has classified it under the provisions of the DOT regulations 49 CFR 100-199

4 2 3 3 **Proposed loading plan (when directed by TDA)** When directed by TDA a proposed loading plan shall be submitted for approval using the format of the dash number sheet standards contained herein Proposed carloading plans for ammunition, explosives, and hazardous materials shall be coordinated by TDA with, and have the approval of the Bureau of Explosives prior to submittal to cognizant systems command headquarters Loading plans for open-top railcars should be coordinated with the originating carrier prior to submittal to TDA

4 2 4 **Testing and design** Unless directed by TDA, specific testing of carloads will not be required if the loading equipment (blocking, bracing staying etc) is in substantial conformity with existing rules and regulations When directed testing of carloads shall conform to the requirements contained in the appendix hereto In preparing specific designs for specific equipment, the format of this standard shall apply

4.2 5 **Design criteria for railcar loading** Design carloading blocking and bracing to withstand car impact speeds of 8 miles per hour with due regard to lateral sway in transit For explosive and dangerous articles the design must be such that the load will not have any movement that might cause damage to the lading during transit

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4.2.6 **Carloading principles** Sound carloading can be achieved in practice only by careful observation of all of the following basic principles

(a) Know the characteristics of the material being loaded and observe all precautions applicable thereto

(b) Use the proper carrier equipment for the material being loaded Specifically, use cars certified for Class A or B explosives when required Make sure the carrier understands the purpose for which the equipment will be used Do not use cars larger than necessary

(c) Examine carrier equipment upon receipt to ensure that it is, in all respects completely suitable for loading the cargo.

(d) Make all repairs necessary to ensure that the equipment is adequate As a matter of general policy, naval activities should insist upon receiving cars for loading explosives which require only minor repair or no repair Defective railcars furnished by carriers shall be reported to the Naval Sea Systems Command (Code SEA 04H7) as required by OP 2165

(e) Use all of the reference documents applicable to the material being carloaded The MIL-STD dash number sheets and section 5 of this standard list the materials which are applicable to specific problems

(f) Follow the specifications and established design principles in order to block and brace the cargo completely and properly

(g) Be sure containers or pallet loads fit tightly against car walls and/or against each other

(h) Provide adequate doorway protection for the car

(i) Do not overload equipment beyond the load limit

(j) Ensure that overall clearance dimensions are within the limits for unrestricted interchange especially for open-top carloads

(k) Close supervision and inspection of the carloading is essential to ensure compliance with all rules and regulations

(l) A representative of the carrier must sign a written acceptance (car certificate) before a shipment of Class A explosives can be made

(m) In transit, railcars are subject to severe longitudinal and lesser side and vertical shock forces These forces are induced while coupling cars, humping them in marshalling

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yards, and in long trains traveling at relatively high speeds. Car blocking must be adequate to restrain the load against any movement relative to the car that might cause damage to the lading during shipment.

(n) Floating loads that move relative to the car proper are not approved for ammunition, explosive, or dangerous article shipments.

4.2.7 **Inert items.**

4.2.7.1 **Single trip.** Inert items being shipped to Government activities for incorporation into assemblies shall be carloaded in accordance with sound practices. Guidance may be found in the AAR Pamphlets 6, 6A, 14, 27, and Circular 42D.

5 DETAILED REQUIREMENTS

5.1 **General.** Ammunition and explosive shipments are initiated in accordance with the procedures established by current area logistics plans, as approved by the Chief of Naval Operations. Shipments of explosives and other dangerous articles shall comply with all applicable requirements of special and general federal regulations controlling the shipping and transportation of these materials, including publications OP 5, OP 2165, OP 2173, and the Department of Transportation (DOT) regulations. In addition to the federal laws governing interstate transportation, each state and nearly all municipalities have laws or ordinances regulating such transportation within their jurisdiction. Shipments shall comply with all these requirements.

NOTE

When planning to move hazardous materials by rail, MIL-HDBK-236 shall be consulted to determine the proper MIL-STD-1325 dash number sheet to use. This document should be studied so that all of its requirements can be met and the proper equipment ordered. Failure to do this can result in undue delays in shipping schedules and increased costs due to improper use of equipment and loading crews.

5.2 **Ordering car.** The car-ordering procedures outlined herein are applicable to all railcar shipments of the Department of the Navy material. Order only a car of the type and capacity required for the specific shipment.

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5 2 1 Type required Order a car of the type required for example a 40-foot single-door boxcar or a 50-ton flatcar Do not order larger cars than required Use DODX cars, Series 29000, if available and suitable for the material being loaded Because these cars are not equipped with roller bearings as required by the Code of Federal Regulations 49 CFR 100-199 they will not be authorized for shipment of Class A or B items after December 31, 1975 Only pin-type DODX cars, Series 29000, shall be used for over-the-road shipments, rack-type cars are for station use only (See OP 1750 for description of pin-type and rack-type cars)

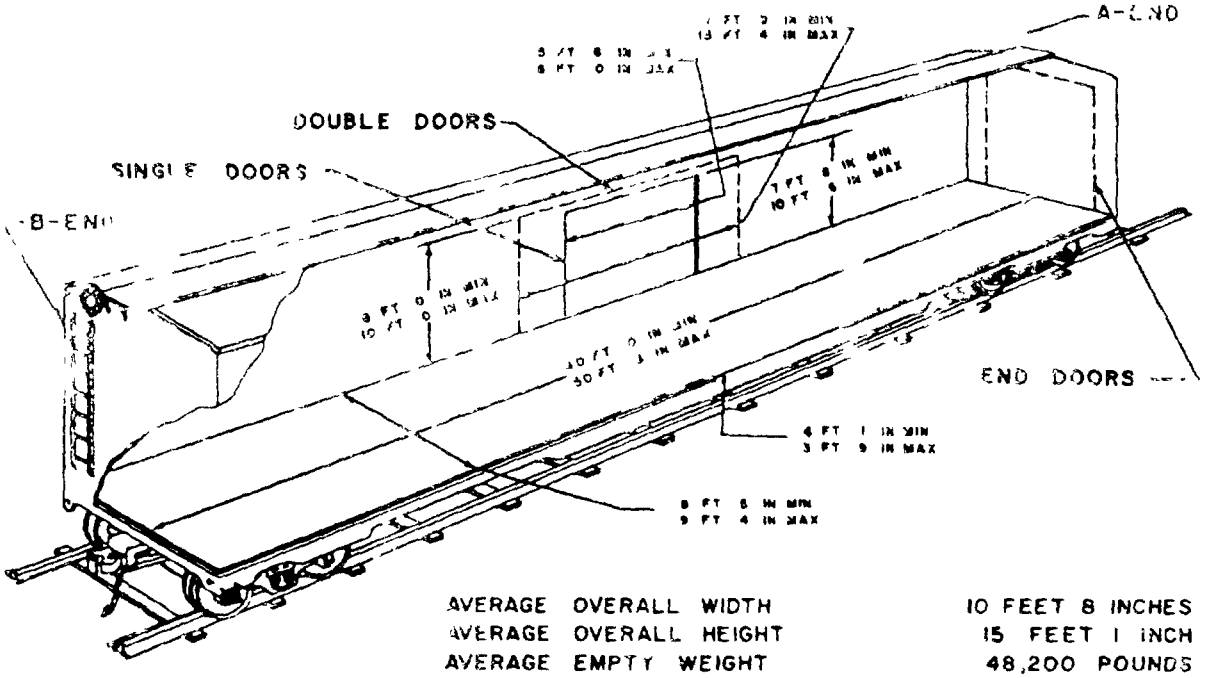
5 2 1 1 On shipments for explosive materials, be certain to stipulate a car suitable for Class A, Class B, or Class C explosives Explosive items suitable for loading in DODX cars Series 29000, are defined in OP 2165 Cars for other dangerous materials shall be suitable for the material being loaded Suitability is defined in detail in sections 174 525 through 174 533 of 49 CFR 100-199 Cars ordered shall be inspected for compliance with the requirements of these regulations before loading Cars which do not meet the requirements of the regulations shall be rejected (For flatcars, see 5 10 2 3)

WARNING

Before loading begins, cars for Class A explosives must be certified as being suitable for use with Class A explosives by certifying representative of the originating carrier Do not begin loading Class A explosives into a railcar unless the car certificate, duly executed by a railroad company employee, as required by section 174 525(c)(3) of 49 CFR Parts 100-199, is held by the shipper, and the loading crew is made aware of the cars certification The shipper shall inspect the car using NAVORD Form 8023/3 Details of the inspection are covered in OP 2165 After inspection of the loaded car two copies of the car certificate must be attached to the car, one to each outer side

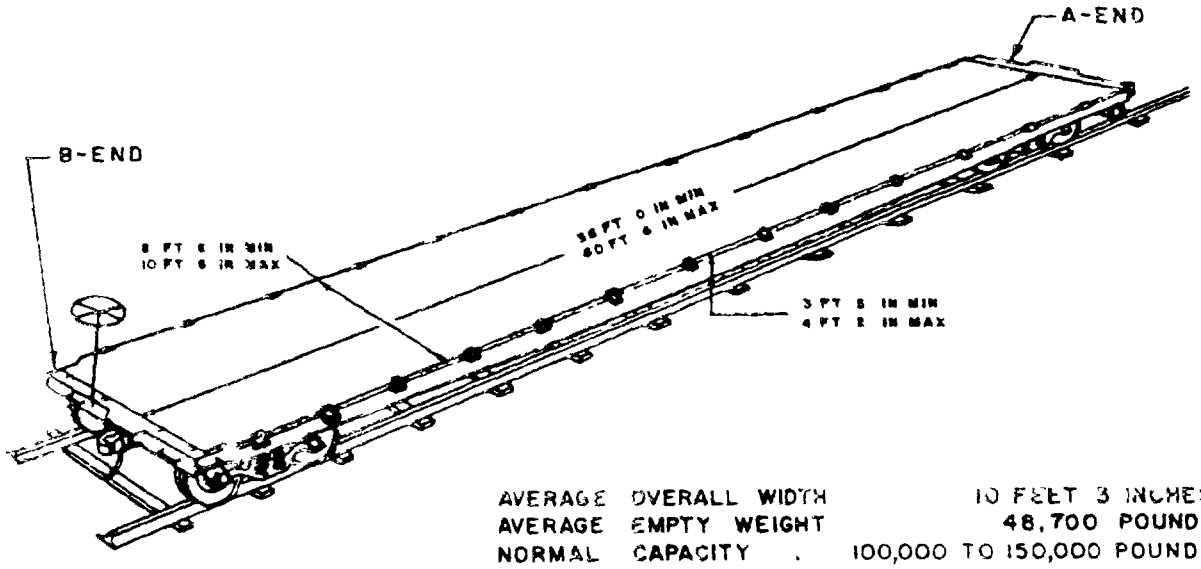
5 2 2 **Size required** Order cars of capacity and size sufficient to carry the shipment authorized Typical car capacities and sizes are shown in figures 5 and 6 Where a minimum car length width, or door width is required, this should be stipulated Maximum clearance dimensions for open car loads are indicated in figure 7 If the proposed lading is over 40 feet long, 8 feet high, and 8 feet wide, contact the nearest military transportation management and terminal agency as required by NAVSUP Publication 444 When rail shipment over foreign roads is contemplated, the dunnaging should be similar but modified to suit the cars used

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AVERAGE OVERALL WIDTH	10 FEET 8 INCHES
AVERAGE OVERALL HEIGHT	15 FEET 1 INCH
AVERAGE EMPTY WEIGHT	48,200 POUNDS
NORMAL CAPACITY	80,000 TO 150,000 POUNDS

FIGURE 5 TYPICAL BOXCAR DIMENSIONS



AVERAGE OVERALL WIDTH	10 FEET 3 INCHES
AVERAGE EMPTY WEIGHT	48,700 POUNDS
NORMAL CAPACITY	100,000 TO 150,000 POUNDS

FIGURE 6 TYPICAL FLATCAR DIMENSIONS

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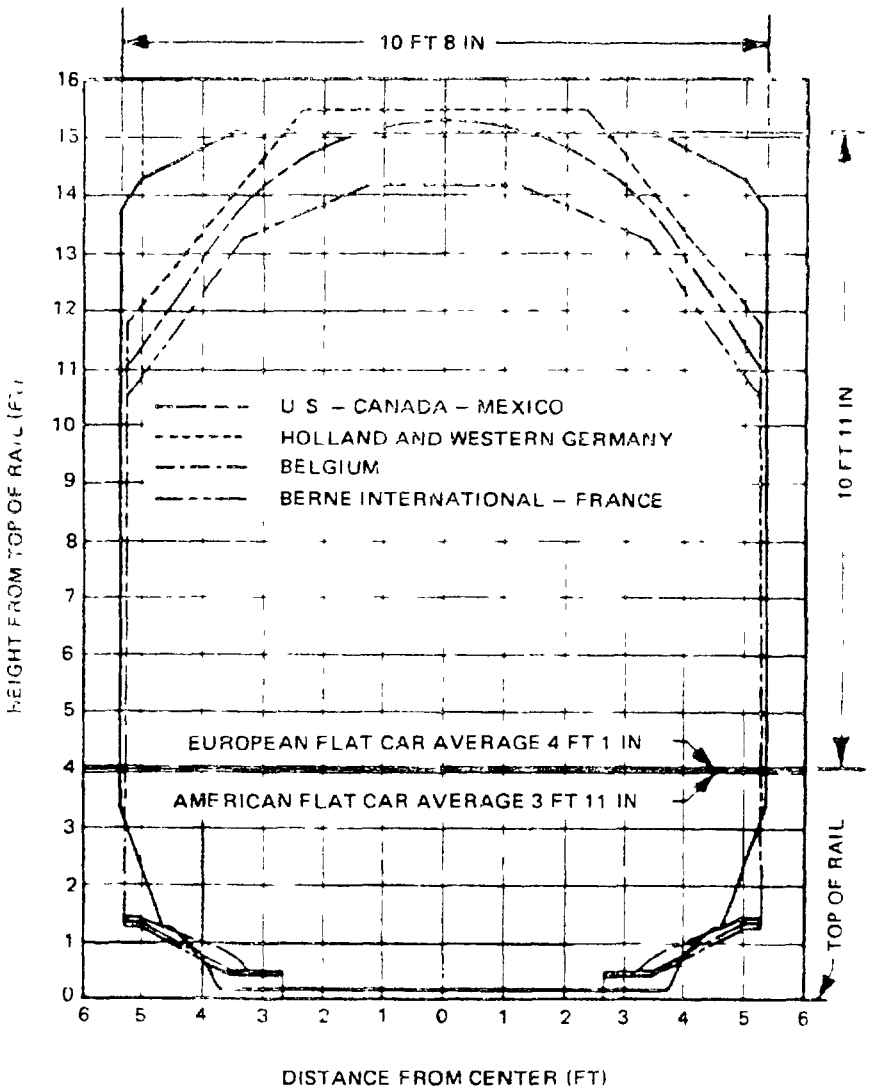
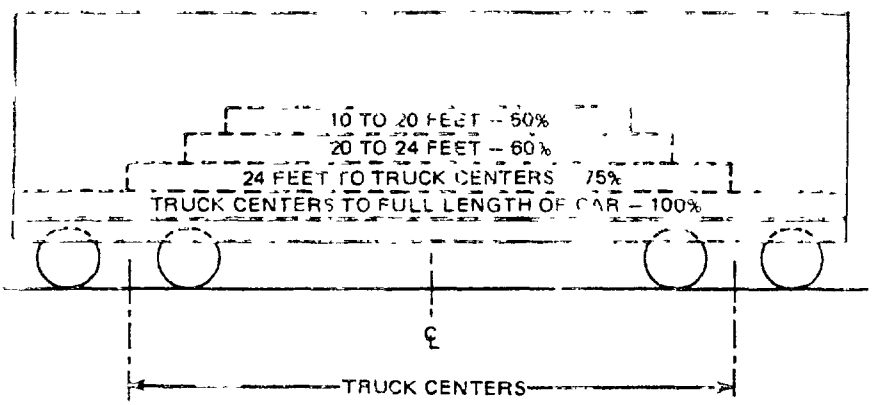


FIGURE 7. MAXIMUM DIMENSIONS OF OPEN CAR LOADS

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[illegible] and weight distribution limitations must adhered to [illegible] figure 8.

BOXCARS OTHER THAN STAGGERED DOUBLE-DOOR CARS BUILT PRIOR TO 1966



FOR STAGGERED DOUBLE DOOR CARS BUILT PRIOR TO 1966

INSIDE LENGTH OF CAR		MINIMUM LENGTH OF LOAD
40 FEET	50 FEET	
40%	30%	20 FEET
50%	40%	25 FEET
75%	75%	25 FEET TO TRUCK CENTERS
100%	100%	TRUCK CENTERS TO FULL LENGTH OF CAR

FLAT CARS WITH BOTH FISH-BELLY CENTER AND FISH-BELLY SIDE SILLS AND ALL FLAT CARS BUILT AFTER 1 JANUARY 1965

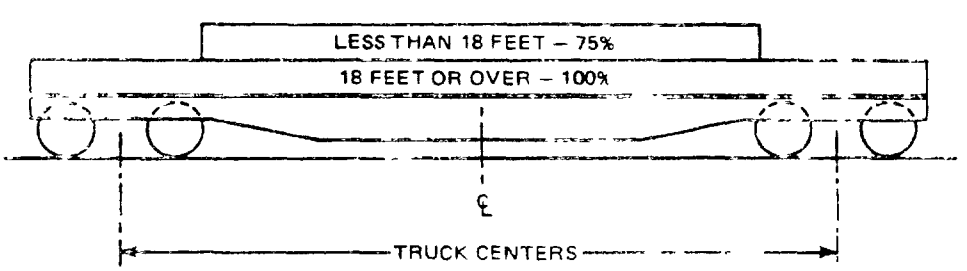
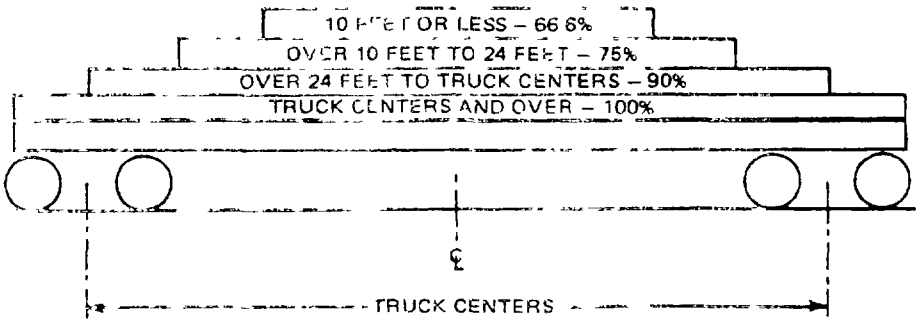


FIGURE 8. MAXIMUM WEIGHT DISTRIBUTION PERMITTED IN CARLOAD CARS

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FLAT CARS NOT EQUIPPED WITH BOTH FISH BELLY CENTER AND FISH-BELLY SIDE SILLS BUILT PRIOR TO 1 JANUARY 1965



GONDOLA CARS

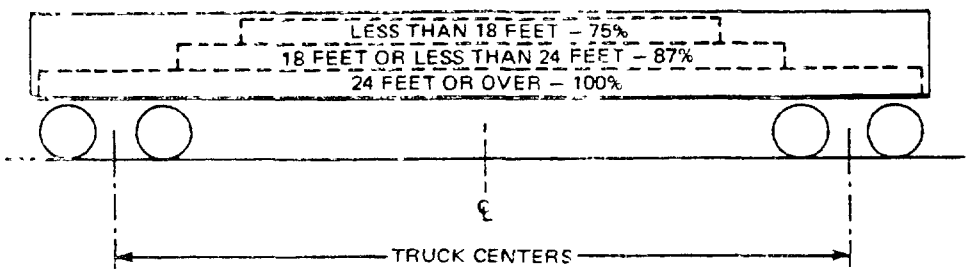


FIGURE 8 (contd)

5.2.2.1.1 The weight of load in or on a car must not exceed the load limit stenciled on car

5.2.2.1.2 The weight of load on one truck must not exceed one-half of the load limit stenciled on car In case of doubt, this should be verified by weighing

5.2.2.1.3 The percentages of stenciled load limits as shown in figure 8 must not be exceeded for loads located between truck centers, measured lengthwise of car unless car owner has otherwise designated by note in the "official equipment register" that these percentages may be changed

5.2.2.1.4 Weight of material loaded in either end between truck centers and end of car must not exceed 15% of stenciled load limit for cars built prior 1 January 1966 and 25% for cars built subsequent 1 January 1966

5.2.2.1.5 For proper distribution of weight crosswise of car, the load must be located so that the weight along both sides of car is about equal for the entire length of the load

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5.2.1.2 Special attention shall be paid to car door size to permit loading of large items such as missile containers. The nominal 40- and 50-foot carloadings shown in the MIL-STD flash number sheets are based on an assumed minimum 6-foot-wide door, unless otherwise specified. Standard 50-foot-long commercial cars have 8- or 10-foot-wide doors while most 50-foot automobile cars have staggered 15-foot doors. DODX cars, Series 29000 have 10-foot, 2-inch-wide doorways.

WARNING

Should less than carload or mixed carload shipments be contemplated, observe the compatibility rules of section 174.538 of 49 CFR 100-199.

NOTE

End-door boxcars shall not be used for shipping explosives and dangerous materials.

5.3 **Inspecting car.**

5.3.1 Before loading operations begin, the car must be checked by the shipper to determine that it is of the type and size ordered and is suitable for the material to be loaded. Boxcars shall have no loose boards or cracks in the floor, roof, sides, or ends through which sparks may enter, nor unprotected decayed spots liable to hold sparks and start a fire. For Class A and Class B explosive, doors shall shut so tightly that no sparks can enter at the joints, and, failing this requirement, the joints shall be stripped with lumber to ensure meeting this requirement. Holes in the floor or lining are not permitted. Particular care shall be taken to ensure that there are no projecting nails, bolts, or exposed pieces of metal which may cause holes to be punctured in containers of explosives during transit. Car floors shall be sufficiently strong to permit operation of loaded forklift trucks within or on the car. Forklift trucks cause a high concentrated load on the floor under their front wheels.

5.3.2 Cars with metal floors, all steel cars, cars equipped with installed bracing devices (including DODX series) and load divider equipped cars shall not be used for shipping bulk explosives that are liable to leakage of dust, powder, or vapor which might become the cause of an explosion. Cars equipped with automobile loading devices shall not be used for shipping explosives. On DODX cars, Series 29000, the loading racks shall be free from any obstructions and be in good working order. The full complement of car equipment shall be present. The number of loader members included as standard equipment in pin-type DODX cars, Series 29000, is as follows:

Wall members	168
Door members	10
Crossmembers	60
Load jack	1

5.3.3 All running gear of the car shall be in good condition. See CP 3651, for inspection procedures.

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5.4 Spotting car at loading area

5.4.1 Move car to loading area. Apply both air and manually operated brakes on the railroad car, or each railroad car of several in a train. Wedge two wheel chocks tightly against one or more wheels placed to prevent movement of the railroad car in any direction.

5.5 Preparing car

5.5.1 Install and secure bridge plate.

WARNING

Check to be sure that bridge plate is secured in place. Unsecured bridge plates can cause casualties.

CAUTION

Make sure that bridge plate is free of ice and grease. In wet or freezing weather, surfaces should be sanded.

5.5.2 Prior to loading, the car shall be swept clean. All protruding nails shall be removed. If this results in loose siding, the siding shall be renailed in new holes. Minor repairs to interior siding, to the extent of replacement of an occasional siding board or plywood panel, may be undertaken if considered desirable in the interest of permitting early shipment. Major car repairs shall not be undertaken. Cars not meeting inspection requirements shall be rejected.

5.5.3 Upon completion of inspection and authorized minor repair, prepare the car for loading operations by installing end wall dunnage or end gates when necessary. End wall dunnage or end gates shall be installed in cars having bowed or unlined corrugated metal ends or when specified in the MIL-STD dash number sheets. All steel boxcars may be used providing bare ammunition does not contact the interior side and end walls of the boxcar. Close door opposite to the one to be loaded, and install prescribed doorway protection for this door prior to placing any material in the doorway area.

5.5.4 When preparing a DODX car, Series 29000, place detachable wall members in positions specified in the MIL-STD dash number sheets. Set aside sufficient crossmembers to meet the requirements of the dash number sheet. Stow unused crossmembers overhead at end of car as required by OP 1750.

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5.6 Loading procedure

5.6.1 Follow the loading procedure given in the MIL-STD dash number sheets and the instructions contained in this section. When a MIL-STD dash number sheet does not exist, follow the loading procedures given in this section.

(a) Begin loading in one end of the car.

WARNING

Use only approved handling equipment when loading ammunition and explosives. See OP 2173.

(b) As soon as the first layer of a stack is in place, install lengthwise sleepers. As additional layers of a stack are placed, install sway bracing.

(c) Make sure containers or pallet loads fit tightly against car walls or against each other, or both as applicable.

(d) After the first stack or bay is in place, position crossmembers or separator gates as applicable. In DODX cars, Series 29000, use the load jack to snug the crossmembers firmly against the load. Basic procedures for operating the equipment in DODX cars, Series 29000, are contained in OP 1750.

(e) Continue loading in the foregoing manner up to the doorway area. Repeat this procedure in the other end of the car. Both ends may be loaded simultaneously.

(f) Complete loading in doorway area as required.

(g) In DODX cars, Series 29000, install doorway members as required and complete installation of crossmembers. If not a DODX car, Series 29000, install center gate structure as required.

(h) In DODX cars, Series 29000, secure all unused equipment in accordance with the requirements of OP 1750.

(i) Install doorway protection when required.

(j) Attach shipping documents to dunnage near door.

(k) Make a complete inspection of all blocking and bracing for exact conformity with the slash sheet requirements. If the load is a Class A explosive, make this inspection in company with a qualified representative of the originating carrier.

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(l) Close doors, lock, and apply strap-type Navy numbered car seals

(m) Apply required DOT placards to outside of car and if a load is a Class A explosive, execute car certificate and attach to car immediately adjacent to the explosive placard

(n) Fill out bill of lading describing the material, using DOT nomenclature as shown in OP 2165

5.7 **Dunnage materials**

5.7.1 The purpose of dunnaging in carloading Department of the Navy material is to prevent longitudinal, lateral, and vertical motion of the lading relative to the car that might cause damage to the lading during shipment When a MIL-STD dash number sheet does not exist, follow the general principles of dunnaging contained in this section When a MIL-STD dash number sheet does exist construct the dunnaging in accordance with the MIL-STD dash number sheet and the general principles contained in this section Modifications in the MIL-STD dash number sheet requirements are permissible only when less than full carload lots are being shipped In this case, activities should follow the principles outlined herein and also those given in the applicable Assocation of American Railroads (AAR) and Bureau of Explosives publications AAR Rules Governing the Loading of Commodities on Open-Top Cars AAR Pamphlets No 14 and 27 AAR Circular No 42E Bureau of Explosives Pamphlets No 6 and 6A

NOTE

The condition of blocking and bracing of any shipment of Class A explosive material shall be inspected and approved by a qualified representative of the Government and the originating carrier

5.7.2 Dunnaging materials in cars which are not special equipped cars consist of lumber, steel nails, spikes, bolts, strapping and seals, wall anchors and occasional pieces of plywood and fiberboard In special equipped cars, dunnaging equipment consists of prefabricated crossmembers which secure into side wall members to retain the lading in bays Some modern commercial cars include cross gates (load dividers) which serve to compartmentize the load They may also have movable side panels which can act as sway bracing These side panels must be fully retracted against the side walls when shipping explosives and, where required, reinforced with appropriate spacers between panel and sidewall In DODX cars, Series 29000, the special equipment is limited to the cross bracing members and associated movable side rails Hence, in numerous cases, it is necessary to use other dunnaging materials to ensure a satisfactory load

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5 7 3 **Lumber** All lumber used shall be yard lumber conforming to MM L-751 or MM-L-736 Unless otherwise specifically indicated lumber used may be rough or dressed Designs are based upon the dressed sizes indicated in table I The species and grades of lumber most commonly used for carloading are listed in table II These wood species are usable for car blocking and bracing in closed cars Where naval ammunition is loaded on open cars, unless otherwise specified, only spruce (eastern Sitka, Tamarack, and white), fir (Douglas) larch (western), hemlock (western), or pine (dense southern yellow, longleaf slash, or loblolly) shall be used

5 7 3 1 **Nominal strengths** Strength values for lumber used for dunnaging are based on past experience as to what values have successfully passed tests or trial shipments, rather than on strictly scientific calculations Strength values for the various species of wood may be found in the Department of Agriculture Handbook 72 In order to standardize drawings, however permitting maximum interchangeability and ability to load cars anywhere in the United States, strength values used in the design of carloading, blocking, and bracing are conservative When selecting the size of lumber for blocking and bracing, consideration should be given to the weight, size, and nature of the material to be secured

5 7 3 2 **Selecting lumber** All blocking and bracing material should be selected from sound lumber free from cross grain dry rot knots knot holes checks, or splits which will affect its strength or interfere with proper nailing Knots, knot holes, checks, and splits or other defects are permitted in lumber as long as they do not impair the strength of the blocking and bracing Blocking and bracing personnel shall take particular care in selecting lumber used in struts, gates cross bracing, side and center bracing, diagonals, holddowns, and K-bracing by upgrading lumber as necessary It is usually possible to upgrade any given piece of lumber by culling through lower grades and, unless the required length is too great cutting out defects (see figure 9)

5 7 3 2 1 The minimum requirement for dunnaging material is common lumber No 2 dimension (exception, southern yellow pine No 3 grade dimension), rough or finished Better grades of lumber will be used only when common No 2 is not available or when used lumber of better grades is available for the same or lower cost

5 7 4 **Fasteners** Fasteners are nails, spikes, and bolts Nails shall be common steel nails conforming to FF-N-105, type II style 10 and shall be bright unless otherwise specified Spikes shall be round wire spikes conforming to FF-N-105 type V style 3 Table III gives actual sizes of nails and spikes

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Table I

SIZES OF DRESSED LUMBER

Nominal dimensions (in.)	Actual dimensions (in.)	
	Softwood	Hardwood
1	21/32	13/16
1-1/4	15/16	1-1/16
1-1/2	1-3/16	1-5/16
2	1-1/2	1-3/4
3	2-1/2	2-3/4
4	3 1/2	3-3/4
5	4-1/2	-[1]
6	5-1/2	[1]

[1] Finished size not specified in grading rules

Table II

SPECIES AND GRADES OF LUMBER REQUIRED FOR CARLOADING

Species	Heavy car blocking	Car bracing	Association grading rules[1]
Softwoods			
Cedar			
Western red	Standard timbers	Standard dimension	WCLIB
Western red	No. 2 timbers	No. 2 timbers	WPA
Cypress	No. 2 timbers	No. 1 common	SCMA NHLA
Douglas fir			
Coast type	Standard timbers	Standard	WCLIB
Mountain type	No. 2 timbers	No. 2 dimension	WPA
Fir			
Balsam	No. 2 timbers	No. 1 dimension	NELMA NPMA
White	No. 2 timbers	No. 2 dimension	WPA
White	Standard timbers	Standard dimension	WCLIB
Hemlock			
Eastern	No. 2 timbers	No. 2 dimension	NHHMA
West Coast	Standard timbers	Standard framing or standard studding	WCLIB
Larch, western	No. 2 timbers	No. 2 dimension	WPA
Pine			
Lodgepole		No. 2 dimension	WPA
Norway (red)	No. 2 timbers	No. 1 dimension	NPMA
Ponderosa		No. 2 dimension	WPA
Southern yellow	No. 3	No. 3	SPIB
Redwood	Snap common timbers	Snap common dimension	CRA

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Table II (contd)

Species	Heavy car blocking	Car bracing	Association grading rules[1]
Spruce			
Engelmann		No 2 dimension	WPA
Eastern	No 2 timbers	No 1 dimension	NELMA NPM
Sitka	Standard timbers	Standard construction	WCLIB
Hardwoods	Hardwood Hearts or common timbers	Hardwood Hearts or No 2 dimension	NHLA

[1]WCLIB West Coast Bureau of Lumber Grades and Inspection WPA Wood Pine Association, SCMA Southern Cypress Manufacturing Association NHLA National Hardwood Lumber Association NELMA Northeastern Lumber Manufacturing Association NPMA Northern Pine Manufacturing Association NHHMA Northern Hemlock and Hardwood Manufacturing Association SPIB Southern Pine Inspection Bureau and CRA California Redwood Association

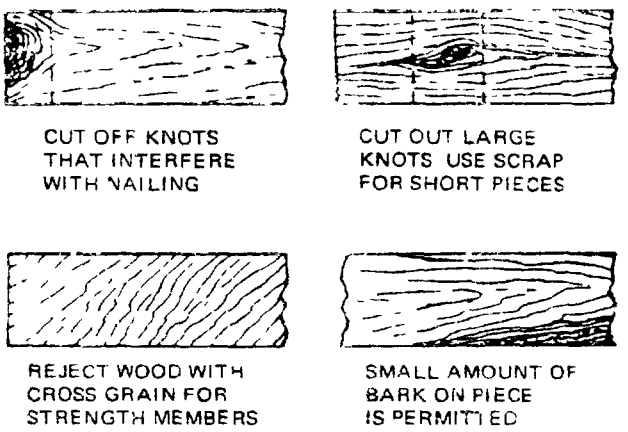


FIGURE 9 LUMBER DEFECTS

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Table III

SIZES OF NAILS AND SPIKES

Size (d = penny)	Nails		Spikes	
	Length (in.)	Diameter (in.)	Length (in.)	Diameter (in.)
2d	1	0.072	–	
3d	1-1/4	0.080	–	–
4d	1 1/2	0.099	–	–
5d	1-3/4	0.099	–	–
6d	2	0.113	–	–
7d	2-1/4	0.113	–	–
8d	2-1/2	0.131	–	–
9d	2 3/4	0.131	–	
10d	3	0.1483	–	–
12d	3-1/4	0.1483	–	–
16d	3-1/2	0.162	–	–
20d	4	0.192	–	
30d	4 1/2	0.207	–	–
40d	5	0.2253	5	0.2625
50d	5 1/2	0.2437	5-1/2	0.2830
60d	6	0.2625	6	0.2830
7 in.	–	–	7	5/16
8 in.		–	8	3/8
9 in.	–	–	9	3/8
10 in.	–	–	10	3/8

5.7.4.1 Nails shall be of such length as to give the necessary holding power and ample penetration into car walls, floors, or other bracing and blocking. To obtain the most holding power, nails shall be of such length that they nearly penetrate but do not protrude through the timber holding the point of the nail. Nails shall not be so large as to cause splitting. The general rule of thumb is that the nail should be three times as long as the thickness of the piece holding the head of the nail, but the nail point should not protrude beyond the second piece unless clinching is required. Recommended sizes consistent with this rule of thumb are given in table IV.

5.7.4.2 All nailing or bolting shall be into the side grain of the lumber, end grain nailing should be avoided. Use plenty of nails. Balanced nailing is important. Stagger nails along the piece being nailed. Do not nail along one grain of wood. Whenever possible drive nails straight; do not toenail unless called for in the MIL-STD dash number sheet.

5.7.4.3 Wooden sheathing of steel frame, single-sheathed cars is usually 1-1/2 to 1-3/4 inch thick, and, in double-sheathed cars, the inside sheathing is usually 3/4- or 7/8-inch thick. Side wall cleats most commonly used are 2- x 4-inch material. Nails driven through

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h[illegible] heathing should not project through the chine thing 8-penny nails should be used for the side walls and [illegible] up by nailing supplemental 1-inch lining using 6d nails, as shown in figure 10. This detail is normally not shown in the MIL-STD dash number sheets.

Table IV

RECOMMENDED NAIL AND BOLT SIZES

Nominal thickness of member holding head (in.)	Nominal thickness of member holding point (in.)					
	1	2	3	4	5	6
1	4d 6d[1]	6d 10d[1]	12d	16d	16d	16d
2	–	10d	20d	40d	40d 60d	40d 60d
3		20d Bolt	40d	60d 6 in	7 in	8 in
4	–	Bolt	Bolt	Bolt or 7 in	Bolt or 8 in	Bolt or 9 in
5	–	Bolt	Bolt	Bolt	Bolt or 9 in	Bolt or 10 in
6	–	Bolt	Bolt	bolt	Bolt	Bolt or 10 in

[1]If clinched

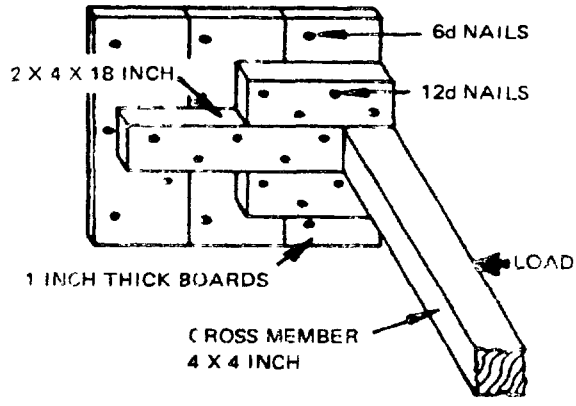


FIGURE 10. NAILING CLEATS TO SIDE WALLS USING SUPPLEMENTAL 1-INCH LINING BOARDS

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5.7.5 Strapping. Steel strapping used in carloading shall be flat strapping conforming to QQ-S-781, type I, heavy duty, finish A, B or C. Heavy duty strapping shall be used in all carloading procedures, whether closed car (boxcar) or open-top car (flatcar), except that regular duty strapping may be used when holding dunnage pieces in position and no load except for the weight of the piece held develops in the strapping during shipment (for example, strapping a buffer board to a DODX crossmember).

5.7.5.1 All heavy duty strapping shall be dry (unwaxed) strapping, and all joints shall be crimped seal joints consisting of two seals (style II, thread-on or closed) each double crimped. Heavy duty strapping, sizes 1-1/4 and 2 inches used for load securements on flatcars, shall be marked to indicate manufacturer's or supplier's name and the letters "AAR" to show compliance with the requirements of the AAR Rules Governing the Loading of Commodities on Open-Top Cars.

5.7.5.2 Regular duty strapping may be lubricated (waxed) strapping, and all joints may be notched or crimped seal joints consisting of one seal (style I, snap-on or open, style II, thread-on or closed, style III, push-type or overlap) double notched or crimped (see figure 11).

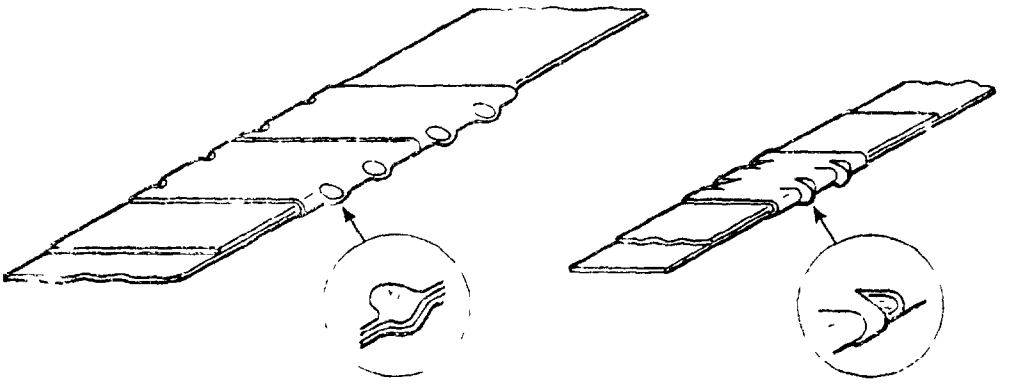


FIGURE 11. TYPES OF STRAP JOINTS.

5.7.5.3 Unless otherwise specified, recommended maximum loads per strap are shown in Table V. Strapping requirements for flatcar loads shall conform to 5.10.4 through 10.4.3.3. Users of tensioning and sealing equipment should be properly instructed in the

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correct use of these tools, and tools should be checked periodically to ensure that the strap seal, and tool combination can produce joints which meet the requirements of QQ-S-781 It is further recommended that hand operated tools be tested not less frequently than once each month

Table V

MAXIMUM LOAD PER STRAP

Strap size (in.)	Load (lb)	Strap size (in.)	Load (lb)
5/8 x 0.020	600	1-1/4 x 0.035	2100
3/4 x 0.025	830	1-1/4 x 0.050	3000
3/4 x 0.035	1260	2 x 0.050	4800

5.5 Dunnaging in DODX cars, Series 29000 The DODX boxcar Series 29000, which is equipped with a pin-type mechanical dunnage system is the only government-owned boxcar authorized for off-Station shipment of hazardous materials After December 31, 1975 these cars will not be authorized for shipment of class A or B items off-Station because they are not equipped with roller bearings as required by the Code of Federal Regulations 49 CFR 100-199 The mechanical dunnage system is designed to eliminate unnecessary damage, eliminate the excessive use of expendable dunnage materials, and reduce loading and unloading time The inside dimensions of the DODX boxcar are length, 50 ft 6 inches between end walls width 8 ft 11-3/8 inches between wall members, and height, 10 feet at the eaves Each car contains, as standard equipment, 168 detachable wall members, 10 doorway members, 60 crossmembers, and 1 load jack In addition, each side wall is equipped with three fixed wall members a series of vertical plates, a series of plywood panels stenciled with height marks, and vertical half plates located in the car doorways (see figure 12) When loading DODX boxcars the load is divided into sections (bays) Each bay, which may contain one or more stacks, is retained by crossmembers Crossmembers must be positioned against strong areas of the lading that are capable of carrying the longitudinal forces Sufficient crossmembers must be used to retain the load in each bay Crossmember capacity for fully distributed loads and loads on the outer thirds of the crossmember is 3000 pounds unless otherwise specified in dash number documents For loads on the center third of the crossmember the capacity is limited to 2000 pounds A check must be made to be sure both ends of all crossmembers are securely locked into the wall members The vertical distance which must be left between wall members to allow room for the installation of crossmembers is shown in figure 13 For more detail description of DODX boxcar dunnage [illegible] procedures refer to OP-1750 MIL-STD-1335-100 and MIL-STD-1335-101 Section 1750 for additional special requirements concerning the use of DODX boxcars

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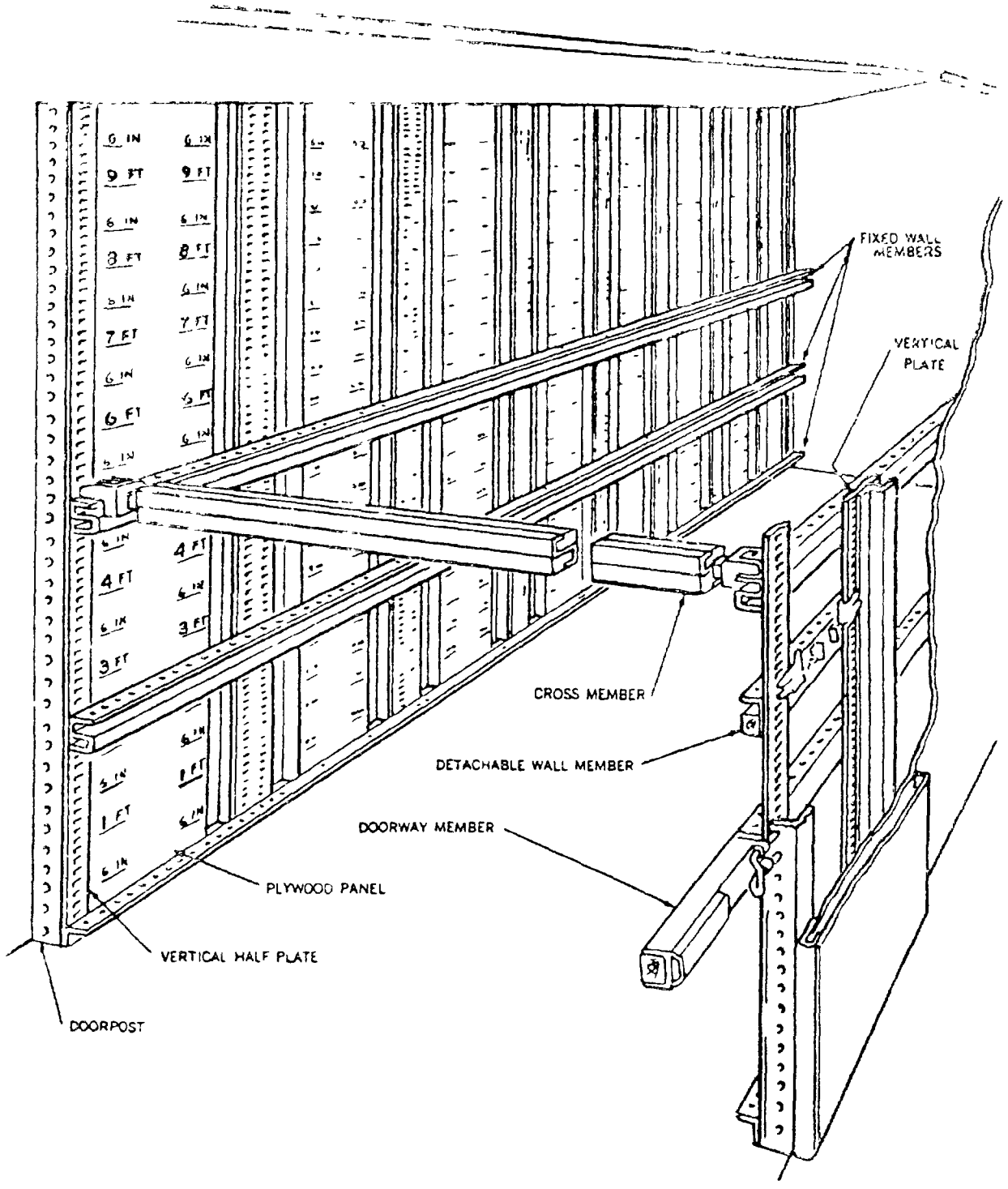


FIGURE 12. PIN-TYPE LOADER SHOWING PARTS

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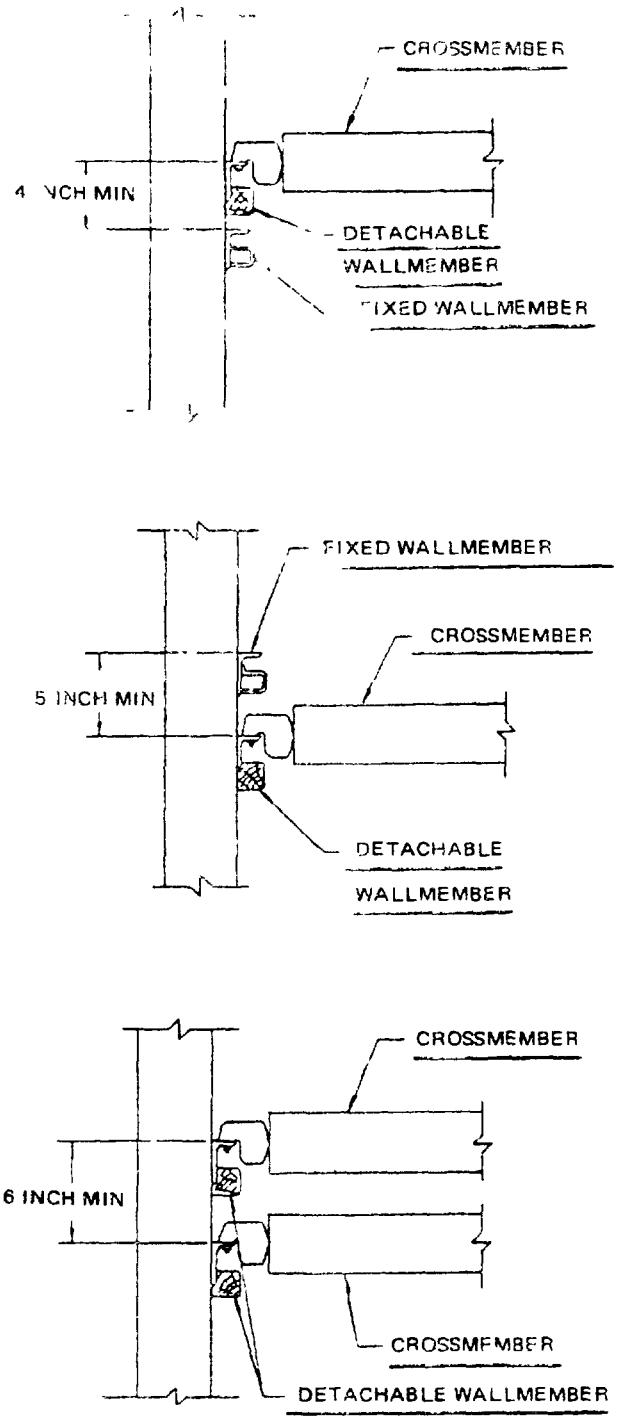


FIGURE 13. VERTICAL DISTANCE BETWEEN WALL MEMBERS

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[illegible] Dunnage [illegible] procedu[illegible] [illegible]ses [illegible] [illegible] blocking and brac[illegible] [illegible] and procedures [illegible] [illegible] are intended to [illegible] [illegible] [illegible] [illegible] individual components of an overall system [illegible] blocking and bracing for a specific type of [illegible] The fundamental concept is to consider the railroad car as an oversized package and to secure the freight in the car so that it is solidly and firmly a portion of the overall package

5.0 It is not practical to provide detailed length dimensions for fabricated cross car components since there is a wide variation in inside car dimensions It is normal practice to allow the shoring crew foremen to order specific lengths to be cut for the specific car being loaded When shipping explosives in all steel boxcars bore ammunition must not contact the interior of the all steel boxcars On certain pallet loads (examples bombs mines or barged propellant charges in tanks) ammunition may overhang the pallet and can contact the interior of all steel boxcars These loads should be shipped in boxcars with interior wood sheathing if possible However if this type car is not available an all steel car may be used provided where ammunition touches steel end walls they are lined with dimensional lumber (minimum 1-inch nominal) or 1/2-inch plywood, and where ammunition touches steel side walls they are lined with dimensional lumber (minimum 1-inch nominal) 1/4-inch plywood 1/8-inch hardboard, or solid fiberboard Solid fiberboard is the most economical for lining side walls and as a minimum requirement must conform to PPP-F-320 type SF (solid fiberboard) class-domestic grade 175 or class weather-resistant grade W6s (0.060-inch thickness) or stronger The lining must be installed in such a manner that it will not shift during transit Lining is not required between metal pallets pallet adapters or ammunition in metal containers and the interior [illegible] cars

NOTE

All cross-section dimensions of dunnage, unless specifically called out otherwise in [illegible] or bills of materials, are nominal dimensions

5.1 End gates End gates are normally used only in cars with bowed or unlined metal [illegible] When end walls are bowed filler pieces and/or shim material must be nailed to the end [illegible] as necessary so that the end gate will bear as much as possible on the bowed end wall A typical installation is shown in figure 14 Sometimes an end gate is necessary for distributing [illegible] loads over the end of the car [illegible] such items as uncrated bombs or empty missiles End wall dunnaging may be nailed to the end wall of the boxcar instead of using an end

5.2.3 Sway bracing Sway bracing is used to prevent lateral movement of elements of the lading as a result of side sway of the car It is used when lading does not completely fill the car crosswise Various forms of sway bracing may be used The most common forms are [illegible] 5.2.3.1 through 5.2.3.4

5.2.3.1 [illegible] or [illegible] [illegible] are members nailed to the car floor after the load is put [illegible] and snugged [illegible] positions They are nailed to the long dimension of the car and [illegible] be of [illegible] 2 [illegible] lumber [illegible] [illegible] skidded containers, sleepers [illegible] [illegible] blocks The use of sleepers is shown in figure 15

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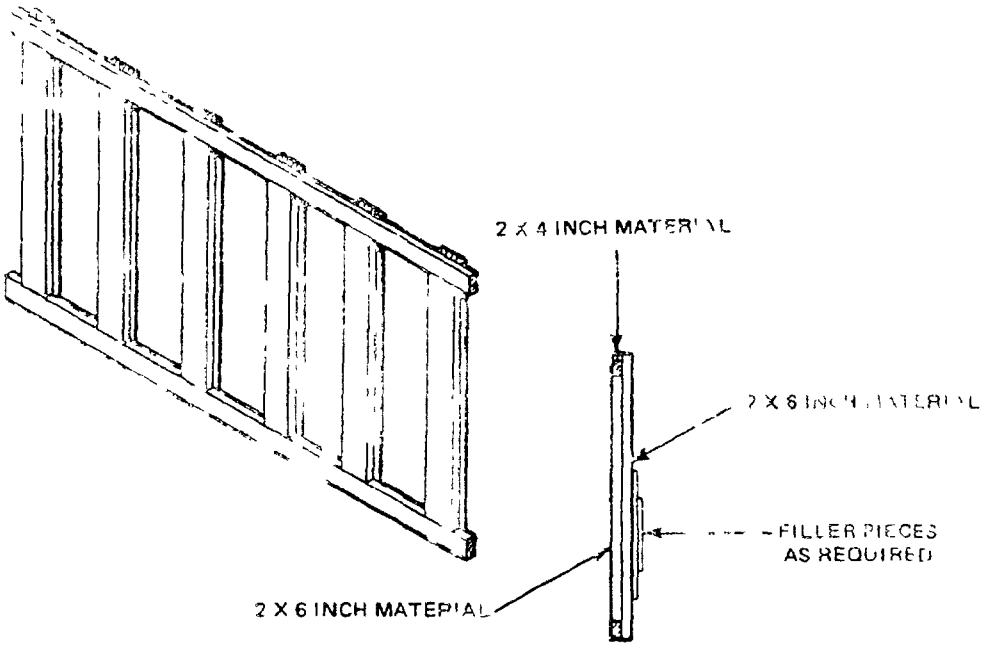


FIGURE 14 END WALL GATE

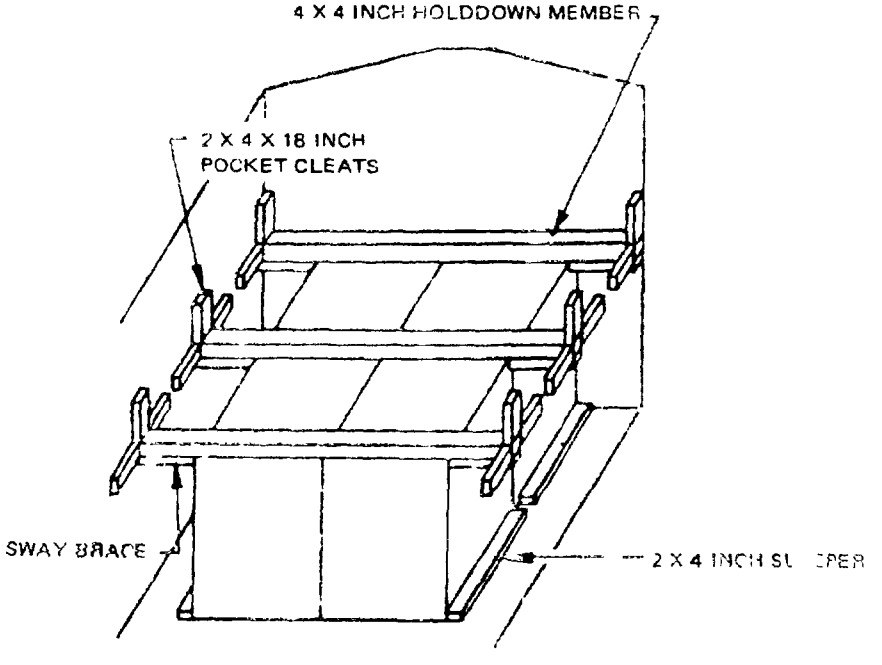


FIGURE 15 HOLDDOWNS WITH TOP-OF-THE-LOAD SWAY BRACING, SHOWING SLEEPERS

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5.9.3.2 **Longitudinal gating** The construction of longitudinal gating is similar to strut-type center gates except that is is placed lengthwise in the car, either along the centerline or along the car wall. Lumber used in the longitudinal gate is usually lighter material than that used in center gates, since crosswise forces are never as great as lengthwise forces in a car. Typical longitudinal gating is shown in figure 16.

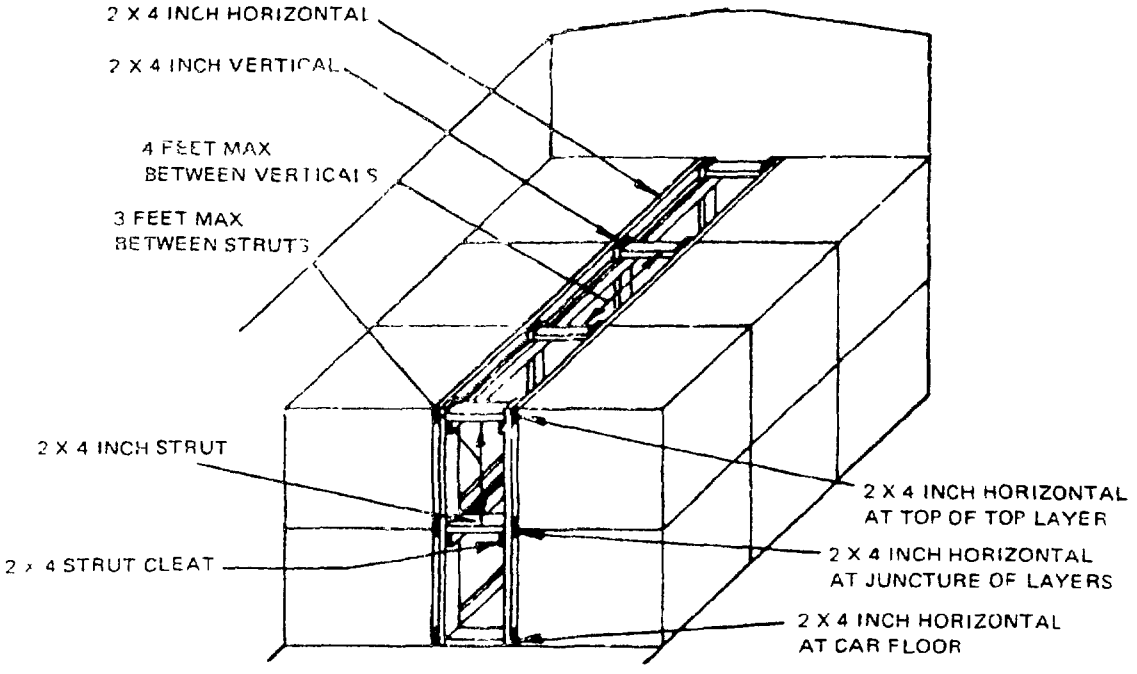


FIGURE 16. LONGITUDINAL GATING

5.9.3.3 **Top-of-the-load sway bracing** These members, nailed or strapped to crossmembers are used to brace the tops of containers or unit loads. Figure 15 shows this type of sway bracing.

5.9.3.4 **Interpallet sway bracing** For sway bracing of pallets in a car, the first layer is braced with longitudinal sleepers nailed to the floor against the pallet. For the second and additional layers, frames are placed between posts of pallets across the car. Stringers and crossmembers are prefabricated to make a frame. The frame is inserted under pallet decks and rest on pallet runners or are nailed to frame supports as shown in figure 17 with one 6d nail at each joint. The width of the frame is 1 inch less than the distance between pallet posts across the car. The frame type sway brace may be used to retain the first layer in cars with steel floors.

5.9.4 **Holddown bracing** Holddown bracing is used to keep loads from displacing vertically and then horizontally. As a general rule, holddown bracing is applied only when a vertical displacement of 3 inches or less could result in vertical or lateral freedom of any element of the load. Typical holddown braces and their fastening to car walls are shown in figures 15 and 18.

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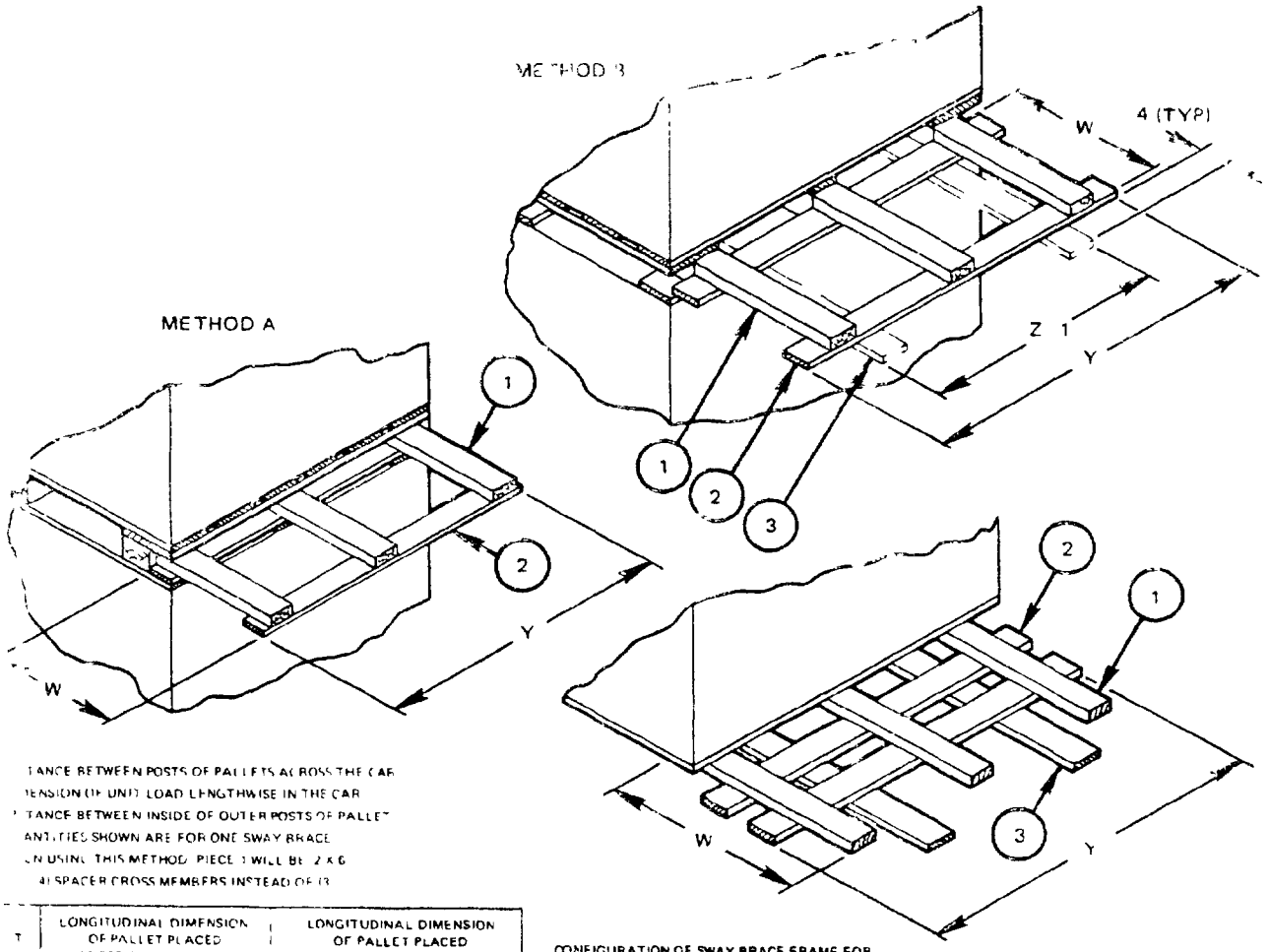
SWAY BRACES

FABRICATION AND INSTALLATION OF SWAY BRACING

THE SWAY BRACE CONSISTS OF A FRAME MADE UP OF ITEMS 1 AND 3 OR OF THE FRAME AND FRAME SUPPORTS (ITEM 3). THE FRAME IS FABRICATED BY NAILING STRINGERS (ITEM 2) TO CROSS MEMBERS (ITEM 1) WITH THREE 10d NAILS CLINCHED EACH JOINT. THE CROSS MEMBERS MUST BE POSITIONED AGAINST POSTS OF ADJACENT PALLETS.

WHEN USING METHOD A, THE FRAME IS FABRICATED AND SLID INTO PLACE BETWEEN PALLETS AS SHOWN. IF CENTER GATE MEMBERS ARE NOT LOCATED SO THAT THEY RETAIN THE FRAMES IN POSITION A SUITABLE LENGTH 2 X 4 MEMBER WHICH WILL RETAIN THE FRAMES MUST BE NAILED TO THE GATE WITH 10d NAILS.

WHEN USING METHOD B, FRAME SUPPORTS (ITEM 3) ARE INSERTED BETWEEN PALLET POSTS AS SHOWN. THE FRAME IS FABRICATED AND POSITIONED BETWEEN PALLETS ON TOP OF FRAME SUPPORTS. STRINGERS (ITEM 2) ARE NAILED TO SUPPORTS (ITEM 3) WITH ONE 10d NAIL EACH JOINT.



TANCE BETWEEN POSTS OF PALLETS ACROSS THE CAR
TENSION OF UNIT LOAD LENGTHWISE IN THE CAR
TANCE BETWEEN INSIDE OF OUTER POSTS OF PALLET
ANTITIES SHOWN ARE FOR ONE SWAY BRACE
ON USING THIS METHOD, PIECE 3 WILL BE 2 X 6
4) SPACER CROSS MEMBERS INSTEAD OF (3

T	LONGITUDINAL DIMENSION OF PALLET PLACED CROSSWISE IN THE CAR	LONGITUDINAL DIMENSION OF PALLET PLACED LENGTHWISE IN THE CAR
12	METHOD A	METHOD B
	METHOD A	METHOD B
	METHOD A	METHOD B
	METHOD A	METHOD B
	METHOD A	METHOD B
39.63 WOOD	METHOD A	METHOD B
1.50	METHOD A	METHOD B

CONFIGURATION OF SWAY BRACE FRAME FOR UNIT LOADS WITH OVERHANG THAT INTERFERES WITH NAILING TO SUPPORT PIECE

PIECE NO.	DESCRIPTION	SIZE	NO. PCS. PER	NAIL TO	NUMBER	SIZE
3	FRAME SUPPORT	1 X 4 X (W - 8)	2	SEE 2		
	SPACER STRINGER	X 4 X Y			SEE NOTE ABOVE	
1	CROSS MEMBER	2 X 4 X W	3	SEE 2		
	LIST OF MATERIAL REQUIRED IN EACH					

FIGURE 17 INTERPALLET SWAY BRACING

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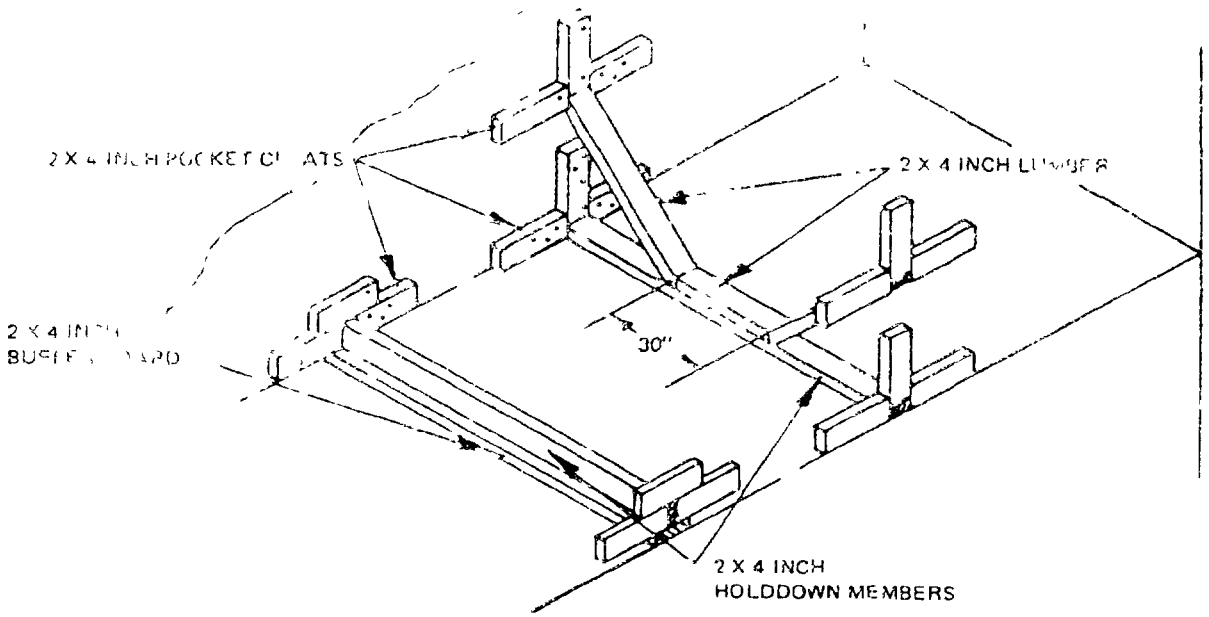


FIGURE 18 HOLDDOWN BRACING

Separator gates Separator gates are used to take up slack in the load, reducing the space required for the center gate or for distributing the load from one bay to the next A typical separator gate is shown in figure 19

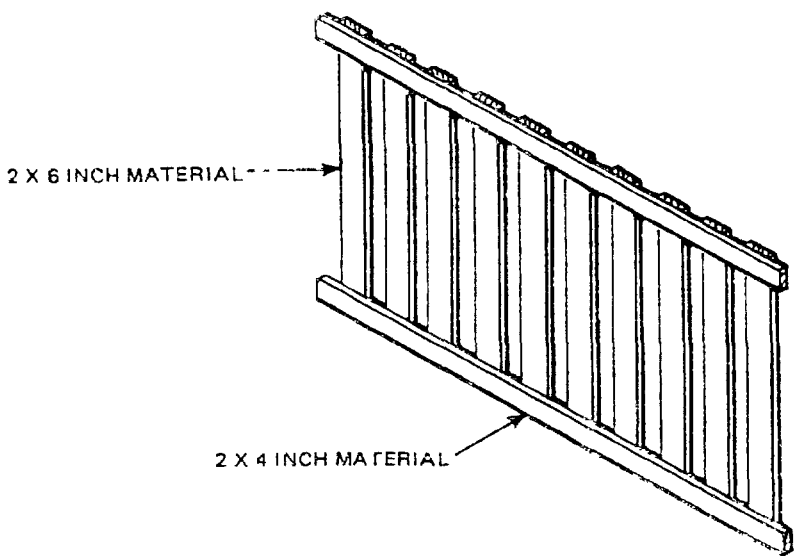


FIGURE 19 SEPARATOR GATE

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5 9 6 Center gates Center gates are used to take up the space in the doorway area of the car to prevent a shift in the load and also to permit the ready removal of lading There are three basic types of center gate structures wedge-type, strut-type, and strut-type with tie bars (see figure 20) When space at center of car is under 20 inches, a wedge-type center gate shall be used providing the height of the gate allows for driving in wedges (approximately 48 inches) Strut-type center gates shall be used when space at center of car is 20 inches or over When struts exceed 48 inches in length horizontal and vertical tie bars shall be added Split gates (one or more rows across car) may be used providing the gates are adequately retained from lateral movement

5 9 6 1 **Verticals** The length of center gate verticals shall normally be equal to the height of the load A minimum of four verticals shall be used, and they shall be located in line with appropriate surfaces (hard spots) of the unit loads, containers, or item comprising the carload For wedge-type gates, outside verticals shall be at least 2 inches in from ends of horizontal members to permit space for tie cleats which are nailed in place last to prevent the wedges from being dislodged

5 9 6 2 **Horizontals** The length of horizontal members of the center gate shall be 1 inch less than the inside width of the car Locate horizontals at or near the top and bottom of the load and in line with appropriate surfaces of the load For palletized unit loads, locate the bottom horizontal 4 inches above the floor For lightly constructed or fiberboard containers, use solid faced gates Nail three 10d nails at each joint of the verticals

5 9 6 3 **Strut cleats** Strut cleats are used to support struts in proper position The length of strut cleats may be 1 inch less than the inside width of car but shall always be long enough to extend past the outside verticals Strut cleats are normally 2- × 4-inch members, but, when struts must be located closer to car floor, 2- × 2-inch members may be used. Strut cleats are normally positioned so that the bottom of the strut will be about 1 inch above the bottom of the horizontal gate members Nail three 10d nails at each joint with the verticals No strut cleats are used on wedge-type gates

5 9 6 4 **Struts** The size, number, and positioning of struts used in a center gate structure shall be carefully determined by the following factors

(a) Locate struts near top and bottom of gate and at or near ends of gates, aligned with horizontals and verticals wherever possible

(b) Intermediate struts should be as equally spaced as alignment permits

(c) Use sufficient struts to distribute load evenly

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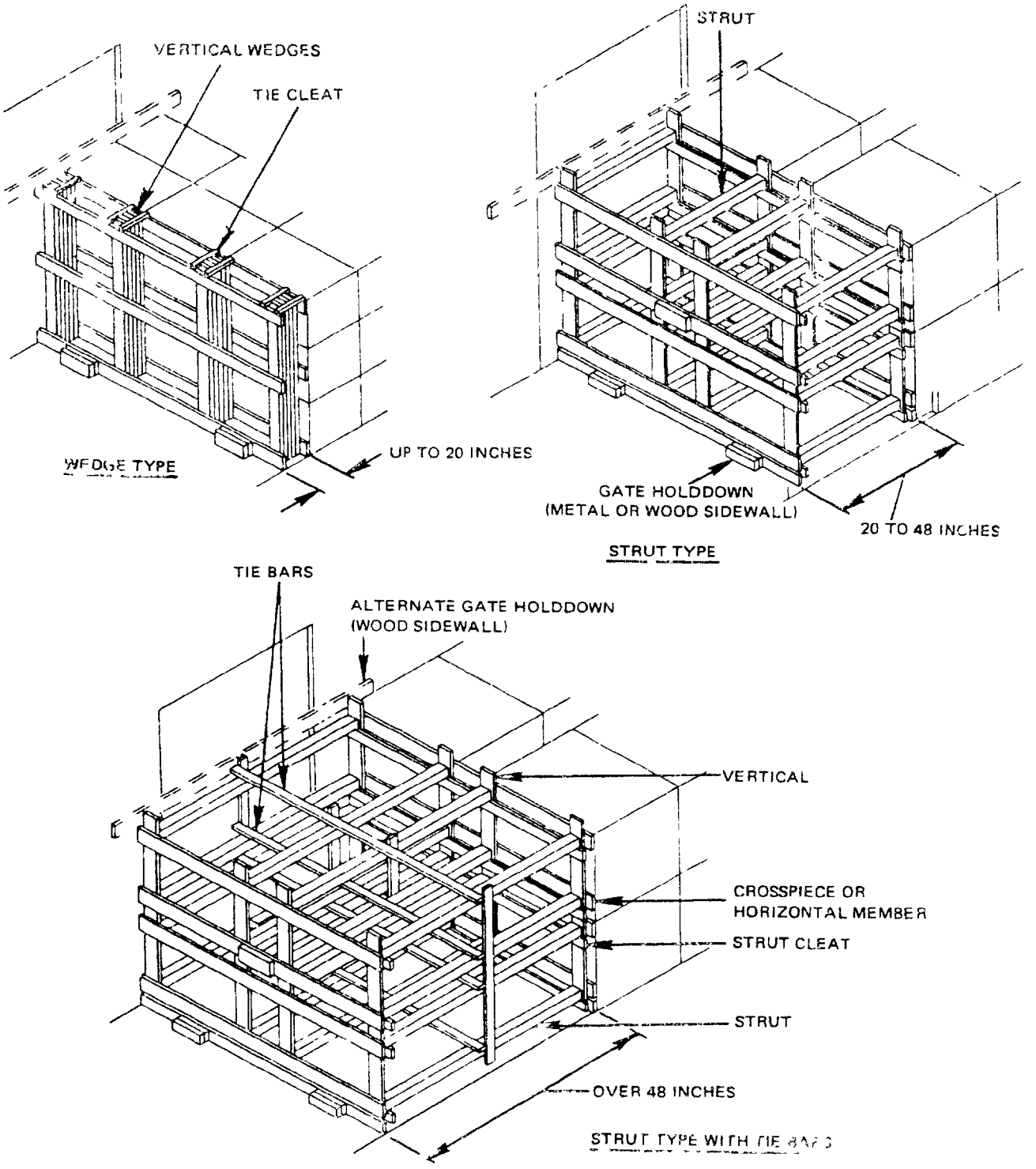


FIGURE 20 TYPES OF CENTER GATES

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(d) Never use less than four sets of struts across the width of the car

(e) Never use less than eight struts

(f) The number and size of struts shall be sufficient so that the load per strut never exceeds the values given in table VI

Table VI

MAXIMUM ALLOWABLE LOAD PER STRUT (POUNDS)

Strut length (feet)	Single 4- x 4-inch strut	Double 2- x 6 inch strut
1	5400	5400
2	5400	5400
3	5400	5400
4	5276	5276
5[1]	5176	5176
6[1]	4998	4998
7[1]	4727	4727
8[1]	4242	4242
9[2]	3585	3585
9½[2]	3163	3163

[1]Struts over 4 feet long must be supported with tie bars at midpoints
[2]Struts over 8 feet long must be supported with two sets of tie bars

5 9 6 4 1 In table VI, the load per strut is obtained by dividing the weight of the load in the heaviest end of the car by the number of struts The strut length is the distance between center gate verticals Lumber sizes are nominal The values in this table assume that tie bars are used to support struts which are more than 48 inches in length

5 9 6 4.2 Struts are normally 4- x 4-inch members Doubled 2- x 6-inch struts may be used in place of single 4- x 4-inch members When installing struts, cut the members slightly longer than the space between gate verticals and hammer in place to make a wedge-tight fit When double 2- x 6-inch struts are used laminate with one 10d nail every 5 inches Toenail to gate verticals with two 12d nails at each end Do not nail struts to car floor or walls

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5 9 6 5 Tie bars When the length of struts exceeds 48 inches horizontal and vertical tie bars shall be nailed at the midpoints of the struts as shown in figure 20 Tie bars are for the purpose of bracing the struts to prevent their springing Nail tie bars to struts with three 16d nails at each joint

5 9 6 6 **Center gate holddown** Center gates are prevented from riding upward and buckling by means of gate holddowns Gate holddowns are principally of two types One type consists of 2 X 6 inch X door width + 48 inch members positioned across the doorway area just above the center gate and bearing on it These members are nailed to the sidewall, each side of the doorway with five 10d nails In addition holddown cleats at least 18 inches long are nailed to the holddown members (five 10d nails per cleat) above the gates to increase the bearing area This type holddown may only be used in boxcars with wood side walls The second type holddown consists of cleats nailed to the gates in such a manner that when finally positioned the center gate assembly is trapped under the lading thereby preventing upward movement relative to the lading The cleats may be doubled or tripled in order that a minimum of 1-1/2 inches are trapped under the lading Each cleat must have at least three 10d nails holding it This type holddown may be used in boxcars with either wood or metal side walls (see figure 20)

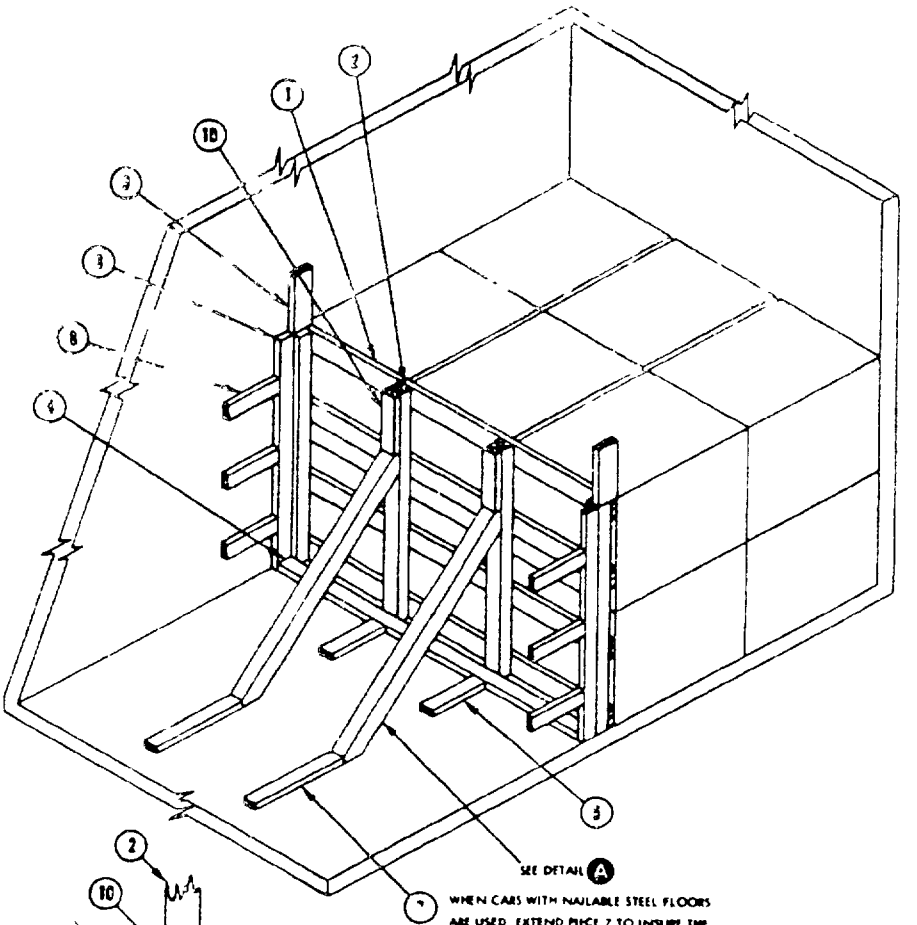
5 9 6 7 **Center gates** Center gates are never nailed to car floors, walls, holddowns, or doorway protection but are left free to move with the load in the event slight shifting occurs

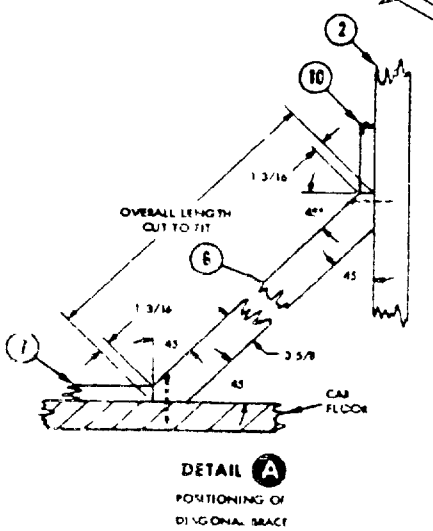
5 9 6 8 When no doorway protection is required, the gate must be prevented from moving laterally against the doors by an appropriate dunnaging method

5 9 6 9 **Limitations** The maximum space to be filled by a center gate structure shall not exceed 9 feet 6 inches Whenever possible, carloads should be designed to avoid exceeding this limitation If this cannot be done, end bracing or partial layer bracing shall be used as described in 5 9 7 and 5 9 8

5 9 7 **End bracing** End bracing may be used for bracing less than carload shipments This method of bracing should not be used in lieu of a center gate structure when the size of shipment will permit the use of center gates Figures 21 and 22 illustrate typical end bracing used for loads not exceeding the weights stated Select sound, straight-grain lumber for diagonal braces The angle which diagonal braces make with the floor shall not exceed 45° nor be less than 30° Adequate nailing is essential to this type of bracing, and good nailing practices must be used When cars with nailable steel floors are used, the floor cleats should be extended to ensure that eight staggered nails can be driven through each floor cleat

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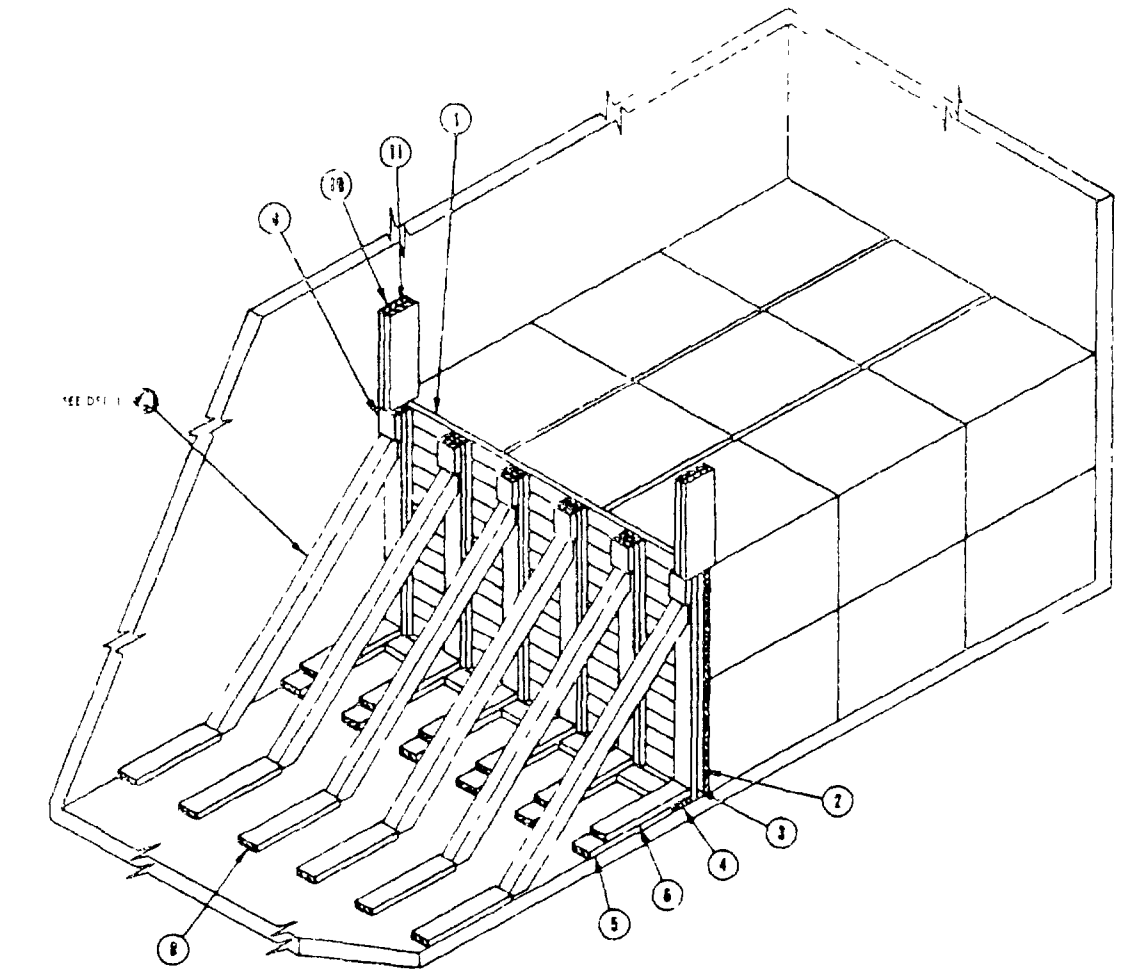
DETAIL A
POSITIONING OF DIAGONAL BRACE

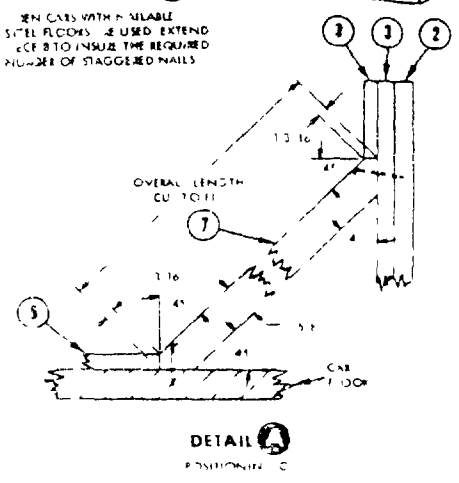
NO	UPPER CLEAT	2 X 4 X CUT TO FIT	2	2	4	10d
9	HOLD DOWN CLEAT	2 X 6 X 18	2	CAR WALL	6	10d
8	SIDE WALL BACK-UP CLEAT	2 X 4 X 18	6	CAR WALL	4	10d
7	FLOOR CLEAT	2 X 4 X 30 MIN	2	CAR FLOOR	5	20d
6	DIAGONAL BRACE	4 X 4 X CUT TO FIT	2	2, CAR FLOOR	2/END	20d
5	FLOOR BRACE CLEAT	2 X 4 X 18	2	CAR FLOOR	4	20d
4	LATERAL FLOOR BRACE	2 X 4 X CAR WIDTH - 3 1/4	1	CAR FLOOR	2/FT	20d
3	VERTICAL BACK-UP	2 X 4 X LOAD HEIGHT	2	CAR WALL	2/FT	10d
2	END BRACE VERTICAL	4 X 4 X LOAD HEIGHT	4	SEE 1	-	-
1	END BRACE HORIZONTAL	2 X 6 X CAR WIDTH	4 MIN	2	3/JOINT	10d
PIECE NO	DESCRIPTION	SIZE	NO PCS REQ'D	NAIL TO	NAILS NUMBER	NAILS SIZE

LIST OF MATERIALS AND NAILING DATA

FIGURE 21 END BRACING FOR LESS THAN CARLOAD SHIPMENTS UP TO 5,000 POUNDS

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PIECE NO	DESCRIPTION	SIZE	NO PC REQ'D	NAIL TO	NUMBER NAILS	SIZE
11	HOLD DOWN CLEAT	2 X 10 X 30	2	10	15	10d
10	HOLD DOWN CLEAT	2 X 10 X 30	1	CAR WALL	15	10d
9	UPPER CLEAT	2 X 6 X 8	6	1		10d
8	FLOOR CLEAT	2 X 6 X 30	6	CAR FLOOR	8	20d
	DIAGONAL BRACE	4 X 4 X CUT TO FIT	6	CAR FLOOR	2 END	20d
6	TOP CLEAT	2 X 6 X 30	6	4	8	20d
	FLOOR BRACE CLEAT	2 X 4 X 30	6	CAR FLOOR	8	10d
4	LATERAL FLOOR BRACE	2 X 4 CAR WIDTH	1	CAR FLOOR	2 EA	20d
3	END BRACE VERTICAL	2 X 4 LOAD HEIGHT			2 EA	10d
2	END BRACE VERTICAL	2 X 4 LOAD HEIGHT		SEE		
	END BRACE HOR ZONTAL	2 X 6 X CAR WIDTH	1 PER LOAD LEVEL		JOIN	10d
					NUMBER NAILS	SIZE

FIGURE 22. END BRACING FOR LESS THAN CARLOAD SHIPMENTS FROM 5 000 TO 30 000 POUNDS

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5.9.8 Partial Layer Bracing. Partial layer bracing of palletized unit loads, skidded unit loads, and unitized containers is required when a layer has a doorway area void greater than the maximum area allowed for center gate assemblies. Partial layer procedures applicable for boxcars with nailable side walls differ from those for boxcars with metal side walls.

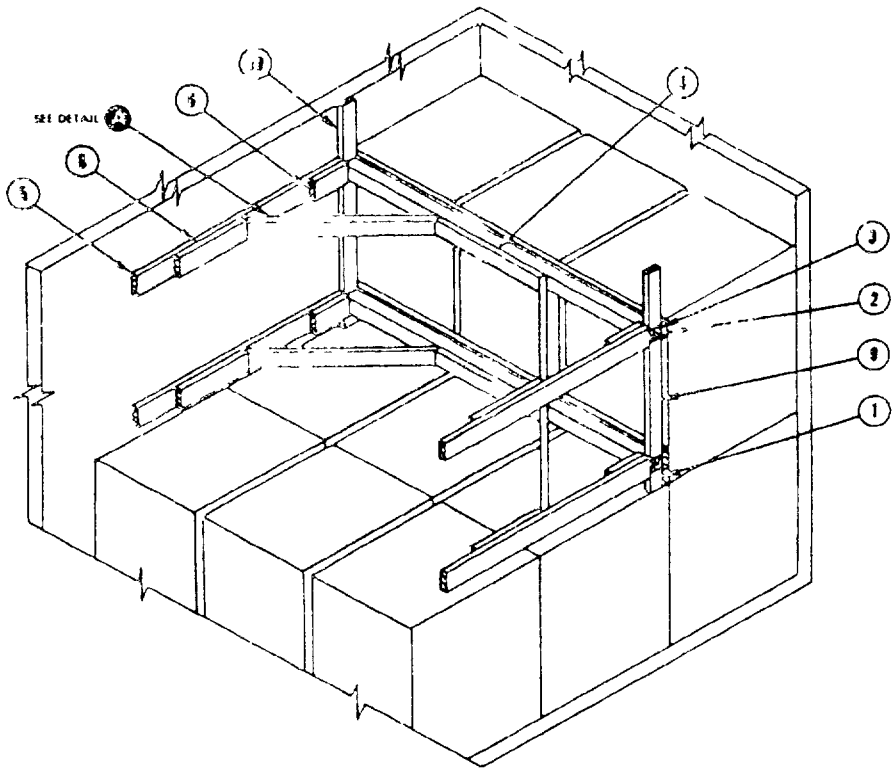
5.9.8.1 Partial layers may be braced with lumber as shown in figures 23 through 26. Diagonal members shall be positioned so that the angle between the wall and the diagonal does not exceed 45° nor be less than 30°. The ends of diagonals shall be double beveled to provide good bearing against bracing members and cleats. Horizontal wall cleats must span a minimum of two side-wall car posts and be secured to the side wall with three nails driven into each post with the remaining nails called for in the nailing data equally spaced. Obviously, this type of partial layer bracing may not be used in boxcars with metal side walls.

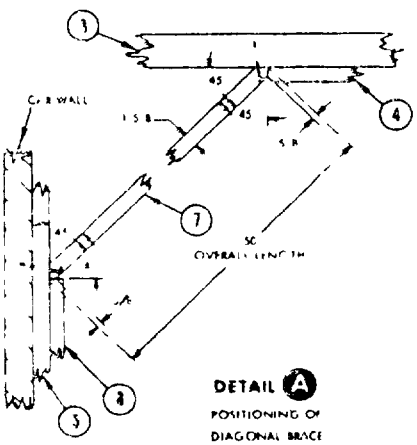
5.9.8.2 Partial layer bracing procedures applicable to boxcars with metal side walls are also applicable to boxcars with wood side walls. The three procedures that have been tested and approved for use are partial layer retention procedures using a unit load on a half-height riser, partial layer retention procedures using knee bracing, and partial layer retention procedures using vertical members and strapping. Typical procedures are shown in figures 27, 28, and 29.

5.9.9 **Floor blocking.** Place floor blocking tightly against the load and nail securely to car floor. Typical floor blocking arrangements are shown in figure 30. Floor blocks are at least nominal 2 x 4 inches. Where the blocking butts against a container skid or load skid, the floor block should be at least as thick as the skid member. Cross blocking should extend the full width of the load and should be provided with backup cleats of the same nominal thickness. Backup cleats should be placed in line with skid members. Backup cleats should extend over three or more floorboards to ensure proper nailing and shall be at least 18 inches long.

5.9.10 **Doorway protection.** Whenever more than half of a pallet or container extends into the doorway area, suitable doorway protection shall be provided. The purpose of doorway protection is to ensure that the load does not damage or jam the door and also to ensure that, even though the door may not be jammed, lading does not fall when the door is opened. Methods of applying doorway protection in cars with wooden doorposts and with steel-jacketed doorposts are shown in figure 31. For cars with steel thresholds, a floor strip shall be cut slightly longer and wedged tightly in place as shown in figure 31. Cross boards shall be spaced on the uprights so that they retain the load from moving laterally in the doorway area.

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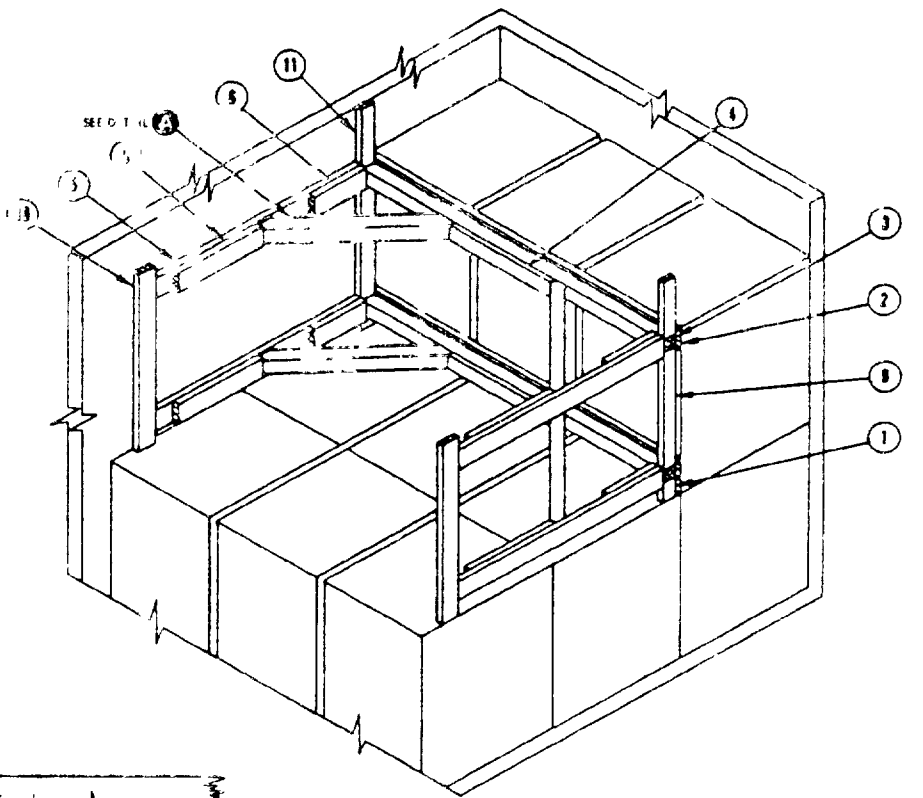


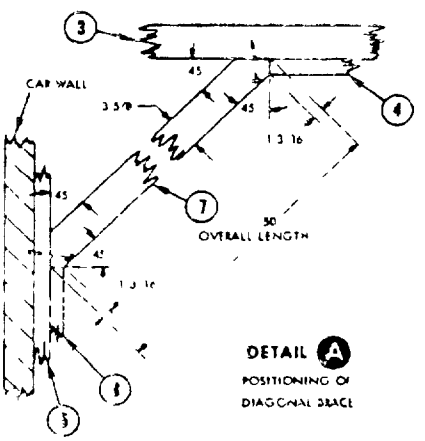


PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQ'D	NAIL TO	NAILS NUMBER	NAILS SIZE
LIST OF MATERIALS AND NAILING DATA						
10	UPPER WALL CLEAT	2 x 4 x 18	2	CAR WALL	4	10d
9	INTERMEDIATE WALL CLEAT	2 x 4 x CLUTCH FIT	2	CAR WALL	4	10d
8	HORIZONTAL BACK-UP CLEAT	2 x 6 x 74	4	5	8	16d
7	DIAGONAL BRACE	2 x 4 x 5[illegible]	4	3	1/END	16d
6	HORIZONTAL POCKET CLEAT	2 x 6 x 1[illegible]	4	[illegible]	4	16d
	HORIZONTAL WALL CLEAT	2 x 6 x 72	4	CAR WALL	16	10d
4	CENTER CLEAT	2 x 4 x 36	2	3	7	16d
3	CROSS BRACE	4 x 4 x CAR WIDTH	2	SEE 2		
2	CROSS BRACE STIFFENER	2 x 6 x CAR WIDTH	2	3	2/FT	12d
1	LOWER WALL CLEAT	2 x 4 x 6	2	CAR WALL	3	10d

FIGURE 23 PARTIAL LAYER BRACING UP TO 8 000 POUNDS

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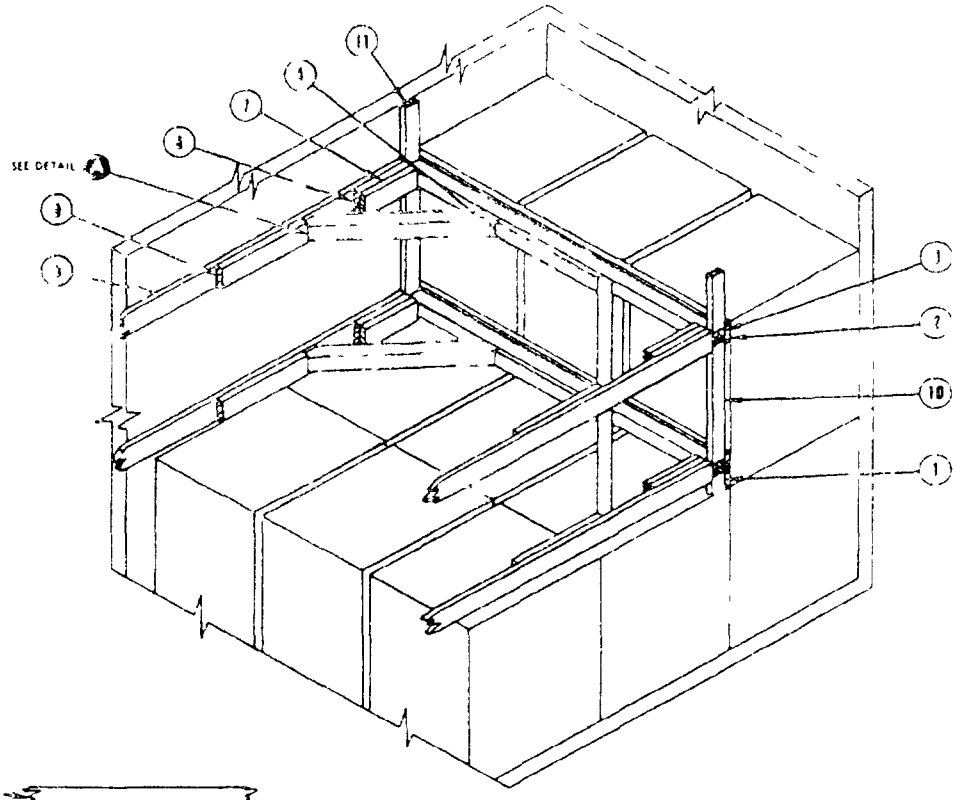
DETAIL A
POSITIONING OF DIAGONAL BRACE

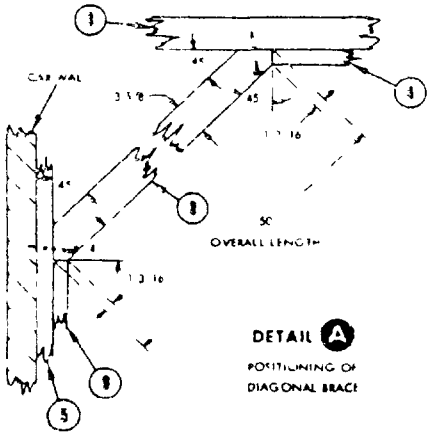
PIECE NO	DESCRIPTION	SIZE	NO. PCS REQ'D	NAIL TO	NAILS NO. PER	NAILS SIZE
11	UPPER WALL CLEAT	2 X 4 X 18	4	CAR WALL	4	10d
10	VERTICAL BACK-UP CLEAT	2 X 8 SLIGHTLY ABOVE LOAD	2	CAR WALL	8	10d
9	INTERMEDIATE WALL CLEAT	2 X 4 X CUT TO FIT	2	CAR WALL	4	10d
8	HORIZONTAL BACK-UP CLEAT	2 X 6 X 30	4	5	14	16d
7	DIAGONAL BRACE	4 X 4 X [illegible]	4	3 *	1/END	60d
6	HORIZONTAL POCKET CLEAT	2 X 6 X 18	4	[illegible]	[illegible]	16d
5	HORIZONTAL WALL CLEAT	2 X 6 X 72	4	CAR WALL	12	10d
4	CENTER CLEAT	2 X 4 X 36	2	3	7	16d
3	CROSS BRACE	4 X 4 X CAR WIDTH	1	SEE 2		
2	CROSS BRACE STIFFENER	2 X 6 X CAR WIDTH	2	3	2/FT	12d
1	LOWER WALL CLEAT	2 X 4 X 6	2	CAR WAL	3	10d

LIST OF MATERIALS AND NAILING DATA

FIGURE 1. PAPER ROLL LAYER BRACING 8,000 TO 14,000 POUNDS

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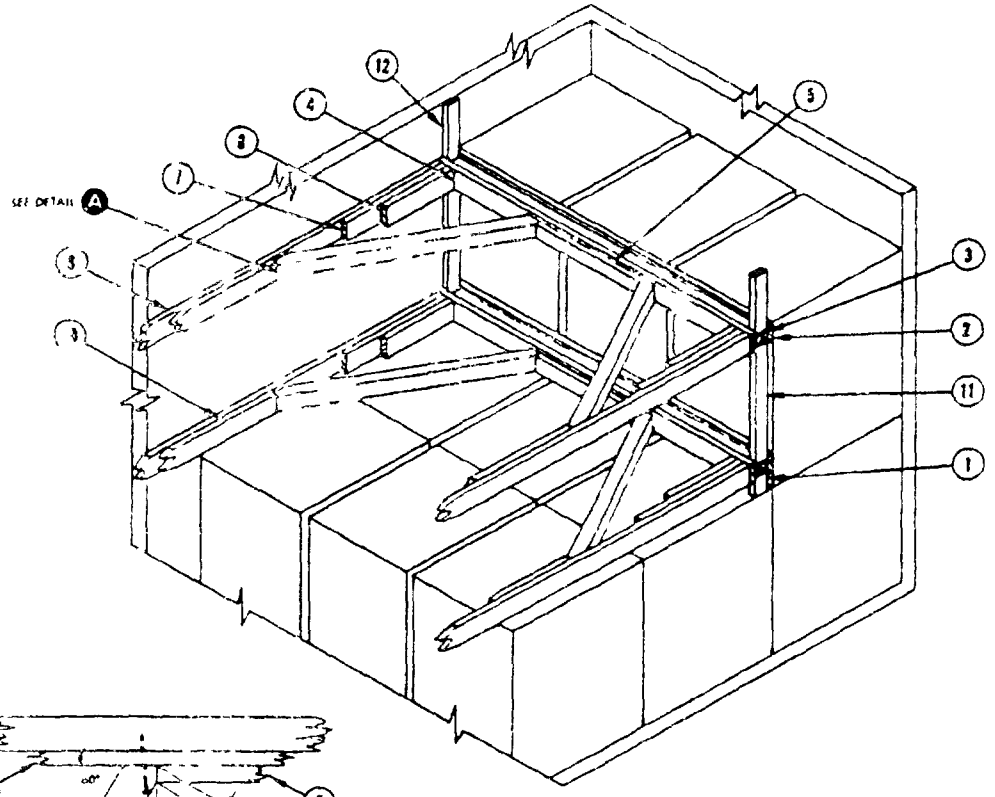


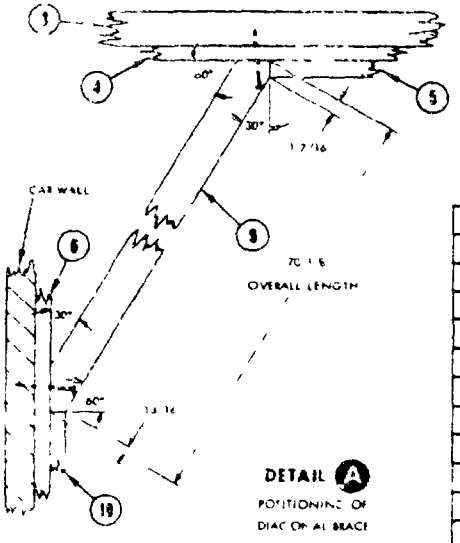
PIECE NO	DESCRIPTION	SIZE	NO. PCS REQ'D	NAIL TO	NAILS NO.-SIZE	NAILS SIZE
11	UPPER WALL CLEAT	2 x 4 x 18	2	CAR WALL	4	10d
10	INTERMEDIATE WALL CLEAT	2 x 4 x CL'G TO FIT	2	CAR WALL	4	10d
9	HORIZONTAL BACK-UP CLEAT	2 x 6 x 30	4		14	16d
8	DIAGONAL BRACE	4 x 4 x 50	4	3	1 END	6od
7	HORIZONTAL POCKET CLEA	2 x 6 x 18	4	6	7	16d
6	HORIZONTAL POCKET CLEAT	2 x 6 x 18	4			10c
5	HOR ZONTA WALL CLEAT	2 x 6 x EXTEND ACROSS DOOR	4	CAR WALL	40	10c
4	CENTER CLEAT	2 x 4 x 36		7		16c
3	CROSS BRACE	4 x 4 x CAR WIDTH	4	SEE 2		
2	CROSS BRACE STIFFENER	2 x 6 x CAR WIDTH	4	3	2 7 1	12c
1	LOWER WALL CLEAT	2 x 4 x 6	2	CAR WALL	3	10d

LIST OF MATERIALS AND NAILING DATA

FIGURE 25 PARTIAL LAYER BRACING 14,000 TO 20 000 POUNDS

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PIECE NO	DESCRIPTION	SIZE	NO PCS REQ'D	NAIL TO	NAIL QUANTITY	NAIL SIZE
12	UPPER WALL CLEAT	2 X 4 X 18	2	CAR WALL	4	10d
11	INTERMEDIATE WALL CLEAT	2 X 4 X CUT TO FIT	2	CAR WALL	4	10d
10	HORIZONTAL BACK-UP CLEAT	2 X 6 X EXTEND ACROSS DOOR	4	6	18	16d
9	DIAGONAL BRACE	4 X 4 X 70 3/8	4	4 6	1 END	60d
8	HORIZONTAL POCKET CLEAT	2 X 6 X 24	4	6		16d
7	HORIZONTAL POCKET CLEAT	2 X 6 X [illegible]	4	5	10	16d
6	HORIZONTAL WALL CLEAT	2 X 6 X EXTEND ACROSS DOOR	4	CAR WALL	40	10d
5	CENTER CLEAT	2 X 4 X [illegible]	2	4	7	16d
4	CROSS BRACE STIFFENER	4 X 6 CAR WIDTH	2	3	2 PT	12d
3	CROSS BRACE	4 X 4 X CAR WIDTH	2	SEE 2		
2	CROSS BRACE STIFFENER	4 X 6 CAR WIDTH	2	3	2 PT	12d
1	LOWER WALL CLEAT	2 X 4 X	2	CAR WALL	3	10d

LIST OF MATERIAL AND NAILING DATA

FIGURE 26. PARTIAL LAYER BRACING 20 000 TO 25 000 POUNDS

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PARTIAL LAYER RETENTION PROCEDURES USING A UNIT LOAD ON A HALF-HEIGHT RISER

THESE PROCEDURES MAY BE USED TO RETAIN AN UPPER PARTIAL LAYER. THE WEIGHT RETAINED IN EACH LAYER ROW MUST NOT EXCEED 8000 POUNDS. THESE TYPICAL PROCEDURES DO NOT APPLY TO ROWS WITH THE SAME NUMBER OF UNIT LOADS RETAINED IN EACH PARTIAL LAYER ROW. THESE PROCEDURES ARE ALSO APPLICABLE TO CARLOADS HAVING OTHER THAN TWO PARTIAL UPPER LAYER ROWS (1, 3 OR MORE ROWS) AND TO CARLOADS HAVING DISSIMILAR NUMBERS OF UNIT LOADS IN THE PARTIAL UPPER LAYER ROWS.

2. CARLOADS THAT DO HAVE DISSIMILAR NUMBERS OF UNIT LOADS IN THE PARTIAL UPPER LAYER ROWS MUST NOT EXCEED THE WEIGHT LIMITATIONS FOR LONGITUDINAL AND CROSSWISE DISTRIBUTION AS REQUIRED BY THIS STANDARD AND THE ASSOCIATION OF AMERICAN RAILROADS (AAR) GENERAL RULES COVERING LOADING OF CARLOAD SHIPMENTS OF COMMODITIES IN CLOSED CARS.

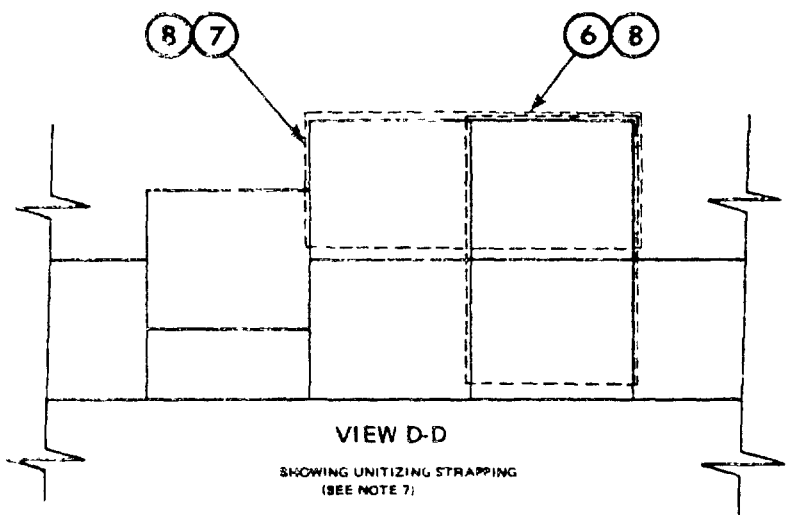
3. EACH STACK MUST HAVE ADEQUATE SWAY BRACING IF THE CROSSWISE VOID EXCEEDS 6 INCHES. CARLOADS HAVING DISSIMILAR NUMBERS OF UNIT LOADS IN THE PARTIAL UPPER LAYER ROWS MUST HAVE THE LONG ROWS SWAY BRACED SO THAT THE LATERAL FORCES ARE CARRIED INTO THE SIDE WALL OF THE CAR (SEE 5.9.3.4 FOR SWAY BRACING PROCEDURES).

4. ONLY APPROVED UNIT LOADS OF ITEMS IN STRONG METAL OR WOOD BOXES/CONTAINERS THAT WHEN PALLETIZED/UNITIZED HAVE A FULLY DISTRIBUTED HARD SURFACE ON AT LEAST TWO OPPOSITE VERTICAL SIDES MAY BE RETAINED BY THESE PROCEDURES (EXAMPLE: SMALL METAL BOXES, WOOD BOXES, ETC).

5. UNIT LOADS MADE UP OF ITEMS WITH IRREGULAR SHAPES AND THOSE THAT REQUIRE PALLET ADAPTERS TO SQUARE UP THE LOAD MAY NOT BE RETAINED BY THESE PROCEDURES (EXAMPLE: PROJECTILES IN DUNNAGE ADAPTER UNCRATED DEPTH CHARGE CASE MK 9).

6. ITEMS THAT ARE PACKED IN BOXES/CONTAINERS THAT MAY ARE NOT STRONG ENOUGH TO CARRY THE LONGITUDINAL FORCES, TO WHICH THESE PROCEDURES SUBJECT THE UNIT LOAD WITHOUT RESULTING DAMAGE TO THE UNIT LOAD MAY NOT BE RETAINED BY THESE PROCEDURES (FIBERBOARD BOXES, BOLSTER CRATES, WIRE BOUND CRATES, ETC).

7. THE TWO STACKS OF THE PARTIAL LAYER ROW NEXT TO THE UNIT LOAD ON THE HALF HEIGHT RISER MUST BE UNITIZED BY STEEL STRAPPING (SEE VIEW D-D). IF ONLY ONE STACK IS BEING HELD BY THE UNIT LOAD ON THE HALF HEIGHT RISER IT MUST BE UNITIZED BY VERTICAL STRAPPING ONLY.



VIEW D-D

SHOWING UNITIZING STRAPPING
(SEE NOTE 7)

*USE 2 X 4 AND 2 X 8 MATERIAL AS REQUIRED TO OBTAIN 1/2 UNIT LOAD HEIGHT ± 2

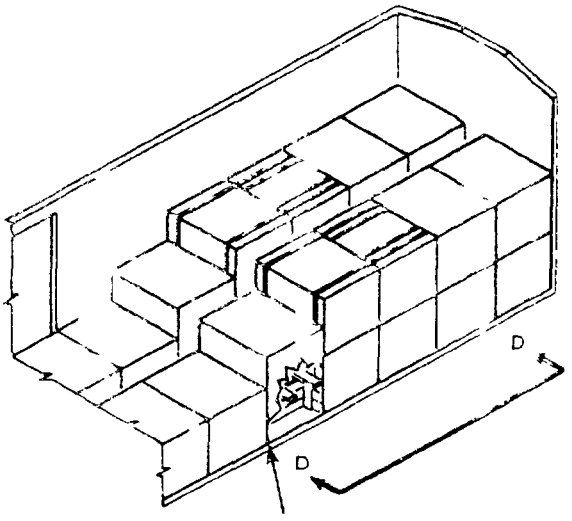
SWAY BRACING OMITTED FOR CLARITY. SEE 5.9.3.4 FOR DETAILS OF SWAY BRACING PROCEDURES.

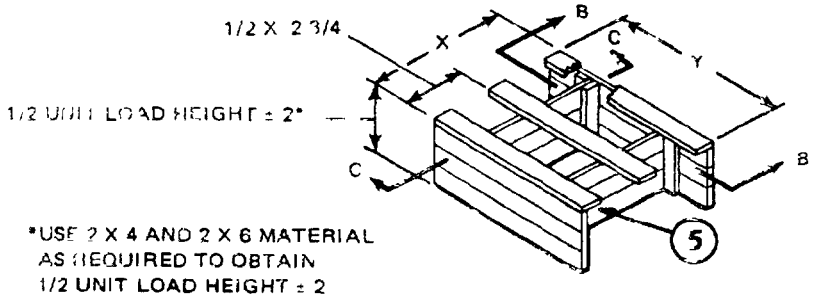
PIECE NO	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NAILS: NUMBER	NAILS: SIZE
8	SEAL	1 1/4	2 PER STRAP			
7	LONGITUDINAL STRAPPING	1 1/4 X .035 X TO SUIT	2 PER ROW			
6	VERTICAL STRAPPING	1 1/4 X .035 X TO SUIT	2 PER ROW			
5	DECK PIECE	2 X 6 X Y	3	3 3	7	10d
4	LONGITUDINAL SUPPORT TIE PIECE	2 X 4 X H	4	3	2 PER JOINT	10d
3	LONGITUDINAL SUPPORT PIECE	2 X 6 X X	SEE NOTE			
2	LATERAL SUPPORT TIE PIECE	2 X 4 X H	4	1	2 PER JOINT	10d
1	LATERAL SUPPORT PIECE	2 X 6 X W	SEE NOTE	3 4	3 PER JOINT	20d

LIST OF MATERIALS AND NAILING DATA

FIGURE 27. PARTIAL LAYER RETENTION UP TO 8 000 POUNDS (Sheet 1 of 2)

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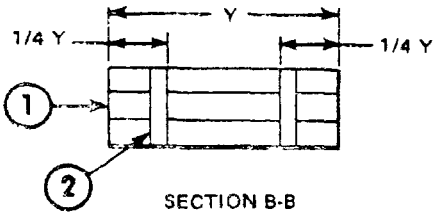
*USE 2 X 4 AND 2 X 6 MATERIAL AS REQUIRED TO OBTAIN 1/2 UNIT LOAD HEIGHT ± 2

X = UNIT LOAD DIMENSION POSITIONED LENGTHWISE IN THE BOXCAR

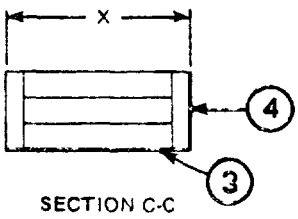
Y = UNIT LOAD DIMENSION POSITIONED CROSSWISE IN THE BOXCAR

DETAIL A

HALF-HEIGHT RISER
1 REQUIRED PER ROW



SECTION B-B
(ITEM 5 OMITTED FOR CLARITY)
LATERAL SUPPORT
2 REQUIRED PER RISER



SECTION C-C
(ITEM 5 OMITTED FOR CLARITY)
LONGITUDINAL SUPPORT
2 REQUIRED PER RISER

FIGURE 24. DUNNAGE FOR 2 LAYER RETENTION UP TO 8,000 POUNDS (Sheet 2 of 2)

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PARTIAL LAYER RETENTION PROCEDURES
U-RODS AND KNEE BRACING

1. KNEE BRACING PROCEDURES MAY BE USED TO HOLD PARTIAL LAYERS HAVING A WEIGHT NOT IN EXCESS OF 24 000 POUNDS EACH, IN OF THE BOXCAR.
2. A MINIMUM OF 1 TIE DOWN STRAP MUST BE INSTALLED FOR EACH LOWER LAYER UNIT LOAD LOCATED BETWEEN THE LOWER ENDS OF THE LONG DIAGONALS EACH END OF THE CAR. MAXIMUM DISTANCE BETWEEN TIE DOWNS MUST NOT EXCEED 48 INCHES.
3. CENTER GATE AREA OF LOWER LAYER SHALL NOT EXCEED 48 INCHES.
4. DIAGONALS, PIECES 17 AND 23 MUST FORM 60° ANGLE WITH THE VERTICAL MEMBER AND A 30° ANGLE WITH THE HORIZONTAL MEMBER (SEE DETAIL A).
5. UNIT LOADS HAVING A HEIGHT GREATER THAN 36 INCHES SHALL HAVE A LONG DIAGONAL HOLDDOWN CLEAT (PIECE 17) 12 INCHES IN LENGTH AND UNIT LOADS HAVING A HEIGHT 36 INCHES OR LESS SHALL HAVE A LONG DIAGONAL HOLDDOWN CLEAT (PIECE 17) 8 INCHES IN LENGTH.

* H - HEIGHT OF UNIT LOAD
** 8 INCHES FOR UNIT LOADS 36 INCHES HIGH OR LESS
*** W - DISTANCE BETWEEN UNIT LOADS IN ADJACENT ROWS

PIECE NO	DESCRIPTION	SIZE	NO PCS REQD	NAIL TO	NUMBER NAILS	SIZE NAILS
25	CROSS BRACE	4 x 4 x W***	AS REQD	SEE 24	-	-
24	SPACER	2 x 4 x W***	AS REQD	25	4	16d
23	SUPPORT PIECE	2 x 4 x (W - 12)***	AS REQD	24	4	16d
				11	2 EACH END	8d
22	SEAL	1 1 4 INCH	2 PER STRAP	-	-	-
21	HOLD DOWN STRAP	1 1 4 x .035 x LENGTH TO SUIT	AS REQD	SEE NOTE 2		
20	STRAPPING BOARD	2 x 6 x (DISTANCE BETWEEN U L +18)	AS REQD	19	5	10d
19	STRAPPING BOARD	2 x 6 x (DISTANCE BETWEEN U L +18)	AS REQD	18	5	10d
18	CROSS BRACE	2 x 6 x CAR WIDTH - 12	AS REQD	16	3 PER JOINT	16c
17	HOLD DOWN CLEAT	2 x 6 x 2**	6	14	5	16d
16	BACK-UP CLEAT	4 x 6 x CUT TO FIT	4	10	2 PER FOOT	10c
15	LONG DIAGONAL BRACE	4 x 4 x 2 (H - 13 .2)*	8	13 & 14	2 EACH END	16d
14	HOLD DOWN CLEAT	2 x 6 x (H - 11 1/2)*	8	2	12	10d

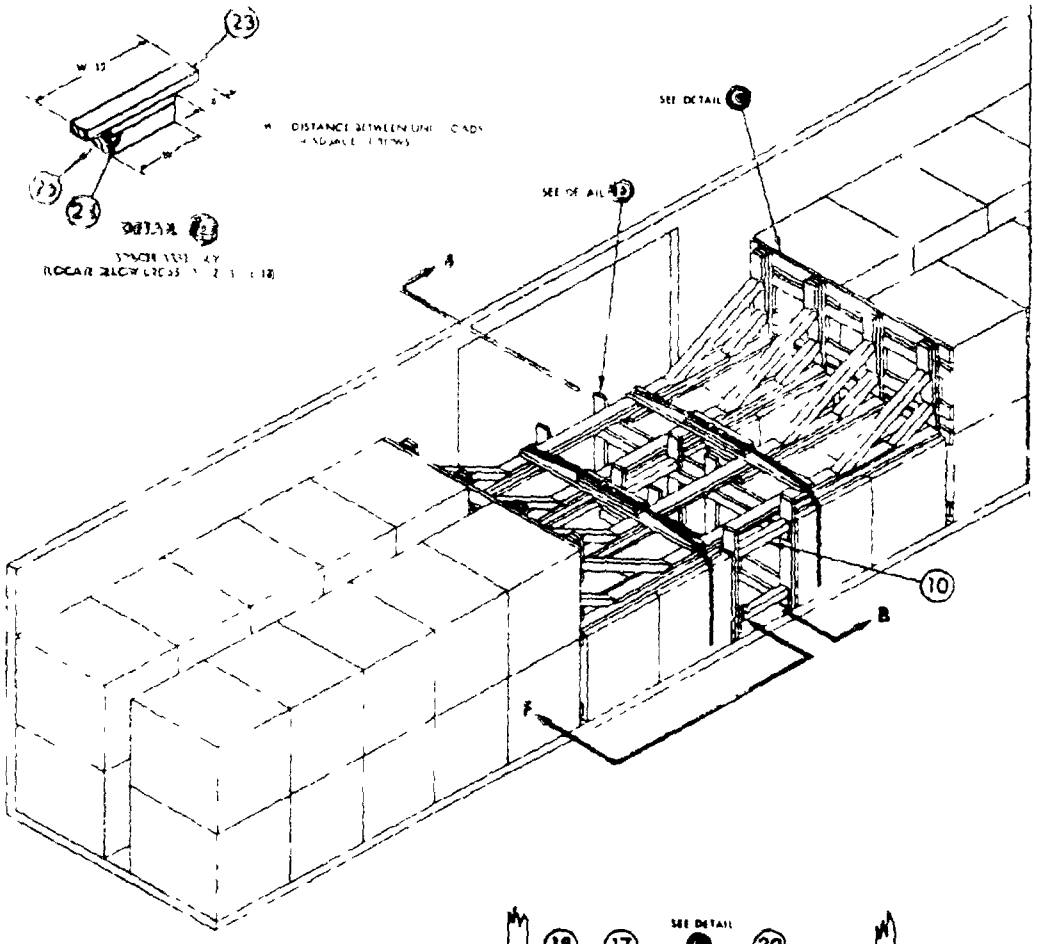
PIECE NO	DESCRIPTION	SIZE	NO PCS REQD	NAIL TO	NUMBER NAILS	SIZE NAILS
13	BACK-UP CLEAT	2 x 6 x CUT TO FIT	4	11	2 PER FOOT	10d
12		4 x 4 x 20	8	4 & 11	2 EACH END	16d
11	BEARING PIECE	2 x 6 x CUT TO FIT	16	SEE 11	-	-
10	STRUT	4 x 4 x WEDGE FIT	8	5	2 PER JOINT	16d
9	CENTER GATE HOLD DOWN	4 x 4 x CUT TO FIT	4	6	2 PER FOOT	16c
8	CENTER GATE HOLD DOWN	4 x 4 x CUT TO FIT	4	-	2 PER FOOT	10d
7	HOLD DOWN MEMBER	2 x 4 x CAR WIDTH - 1	2	5	3 PER JOINT	10d
6	CENTER GATE STRUT CLEAT	4 x 4 x CAR WIDTH - 1	4	5	2 PER JOINT	10d
5	CENTER GATE VERTICAL	2 x 6 x (H - 10)*	8	SEE 4	-	
4	CENTER GATE HORIZONTAL	2 x 6 x CAR WIDTH	4	5	3 PER JOINT	10d
3	SPACER	2 x 6 x CUT TO FIT	8		2 PER FOOT	10c
2		2 x 6 x 2 H		SEE		
1	KNEE BRACE GATE HORIZONTAL	2 x 6 x CAR WIDTH - 1 2	10	4	3 PER JOINT	16c

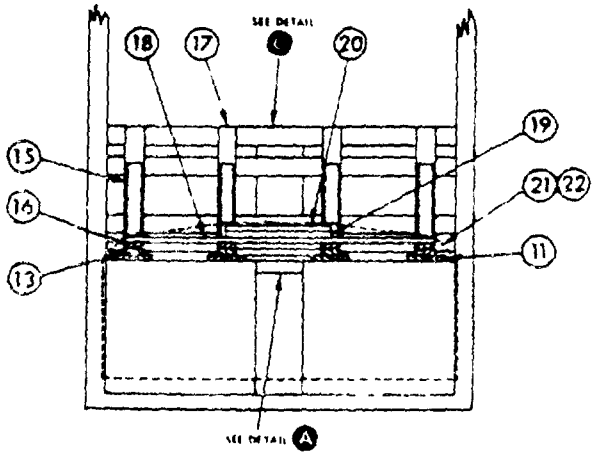
LIST OF MATERIALS AND NAILING DATA

LIST OF MATERIALS AND NAILING DATA

FIGURE 28. PARTIAL LAYER RETENTION UP TO 24 000 POUNDS (Sheet 1 of 3)

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SECTION B-B

METHOD 5013.1
SLING LEG TENSION UP TO 24,000 POUNDS (Sheet 2 of 5)

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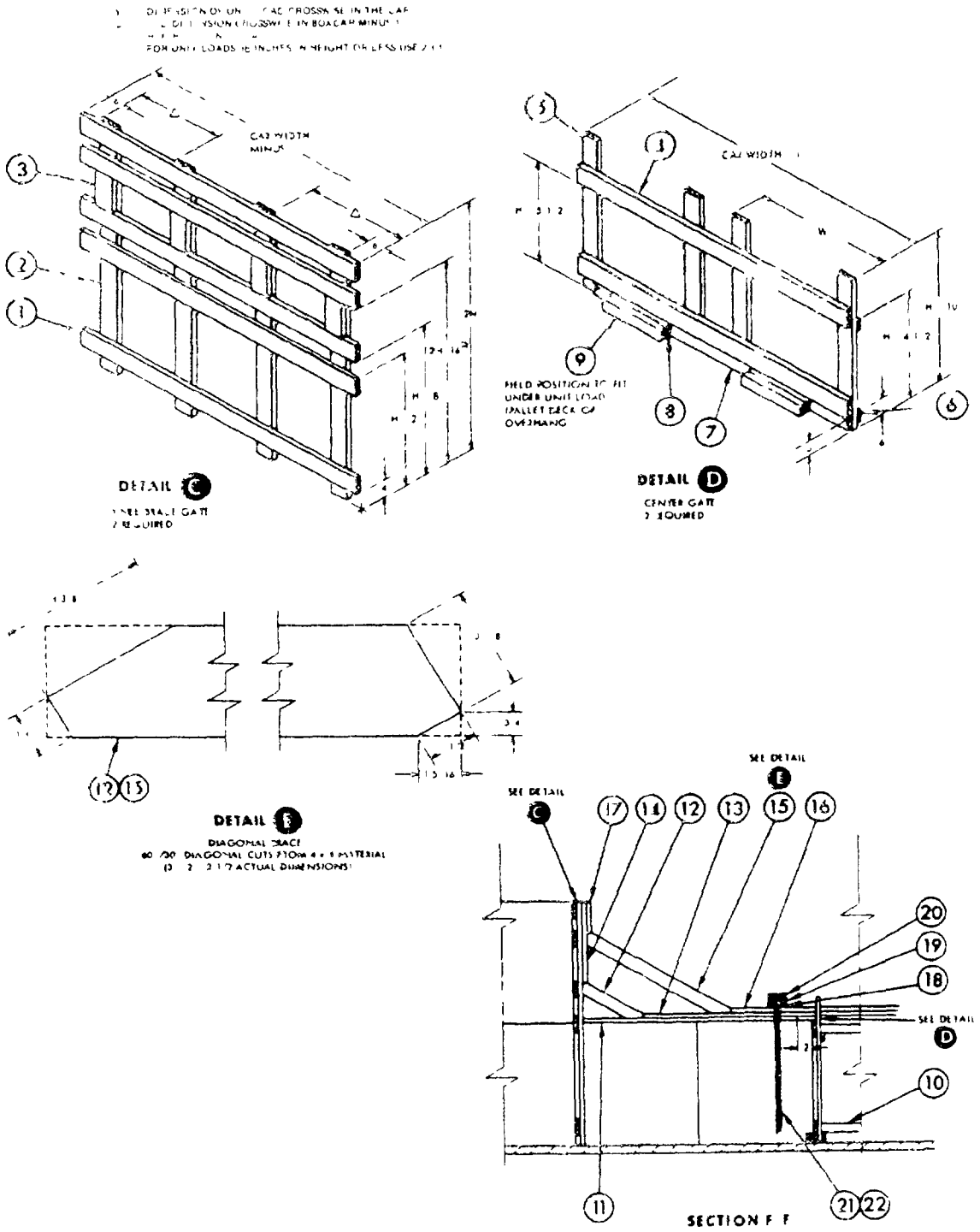


FIGURE 28. PARTIAL LAYER RETENTION UP TO 24,000 POUNDS (Sheet 3 of 3)

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PARTIAL LAYER RETENTION PROCEDURES USING VERTICAL MEMBERS AND STRAPPING

1. THE WEIGHT AND HEIGHT OF THE UPPER PARTIAL LAYER TO BE RETAINED DETERMINES THE SIZE AND QUANTITY OF VERTICAL MEMBERS REQUIRED TO RETAIN THE LOAD. THE AMOUNT OF LOAD WHICH MAY BE RETAINED BY EACH VERTICAL DEPENDS ON THE DIMENSIONS OF THE VERTICAL AND THE HEIGHT IT EXTENDS ABOVE THE TOP OF THE FULL LOWER LAYER. THIS HEIGHT WILL USUALLY BE THE HEIGHT OF THE UNIT LOAD BEING RETAINED. A MINIMUM OF TWO VERTICALS MUST BE USED FOR EACH UPPER LAYER ROW BEING RETAINED. THE TABULATION BELOW SHOWS THE AMOUNT OF LOAD WHICH MAY BE RETAINED BY EACH VERTICAL USED. THE WEIGHT RETAINED IN EACH ROW OF UPPER PARTIAL LAYER MUST NOT EXCEED 8000 POUNDS.
2. A PARTIAL LAYER RETAINING FRAME IS CONSTRUCTED BY NAILING BUFFER MEMBERS TO THE VERTICALS TO DISTRIBUTE THE LONGITUDINAL FORCES OVER STRONG AREAS OF THE UNIT LOADS CONTACTED BY THE FRAMES. THE AMOUNT AND TYPE OF BUFFER MEMBERS REQUIRED DEPENDS ON THE CRUSHING STRENGTH OF THE UNIT LOAD AND ITS ABILITY TO CARRY LONGITUDINAL FORCES.
3. TWO 1/8 INCH STRAPS ARE PLOUGHED TO HOLD A PARTIAL LAYER RETAINING FRAME IN PLACE. THE CONFIGURATION AND HEIGHT OF LOAD DETERMINES HOW AND WHERE STRAPS ARE POSITIONED.

CAUTION

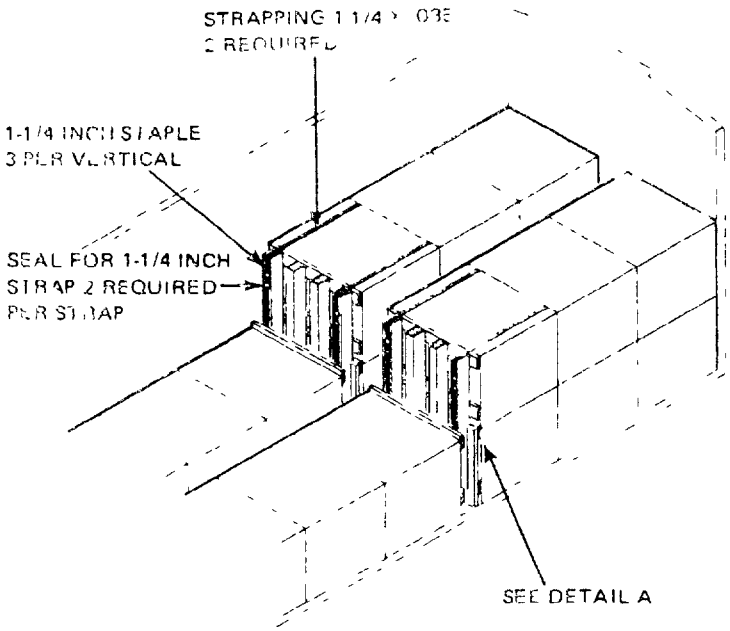
UNIT LOADS THAT ARE WEAK IN AREAS WHERE THE VERTICALS JUST BE POSITIONED AND THAT CANNOT BE SUFFICIENTLY STRENGTHENED BY ADDING BUFFER BOARDS (SUCH AS CRATES, THE BOXED CRATES, POLYSTYRENE BOXES ETC.) AND UNIT LOADS OF AMMUNITION ITEMS (CARTRIDGE CASES, TANKS) SHOULD NOT BE RETAINED IN THIS MANNER.

Height of unit load (in.)	Maximum load per vertical (lb)	
	4 x 4	4 x 6
27	1745	3492
30	1570	3143
33	1428	2857
36	1309	2619
39	1208	2418
42	1122	2245
45	1047	2095

[1] Two 2 X 6 verticals laminated with 10d nails (two per foot) and with a 1 X 4 face board nailed to the outer edge with 6d nails (one per foot) may be substituted in place of 4 X 6 verticals.

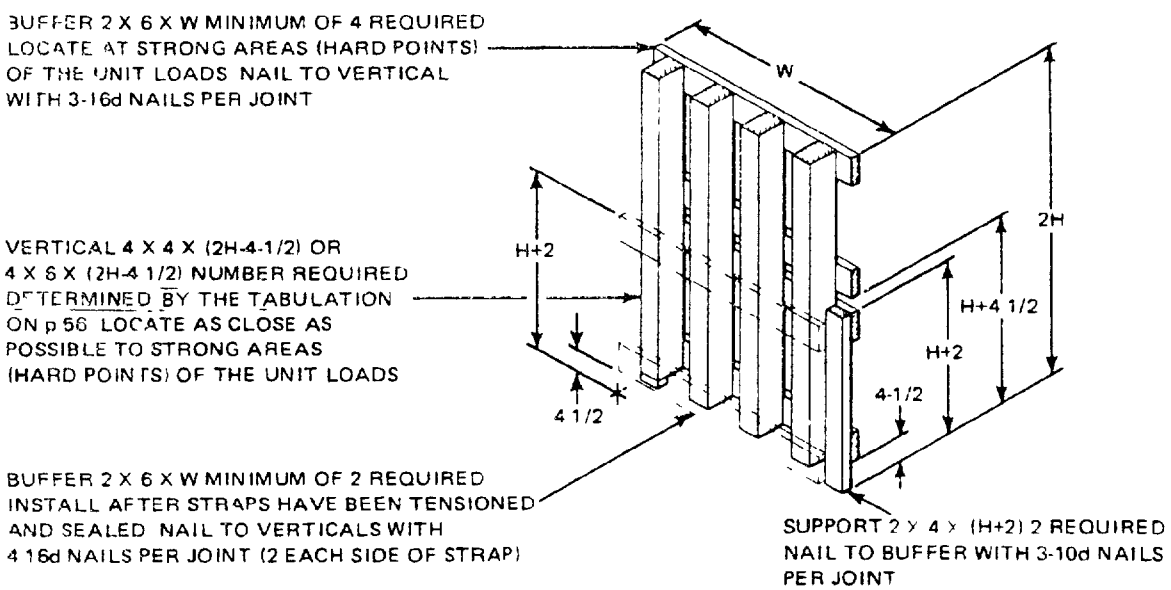
FIGURE 29. PARTIAL LAYER RETENTION WITH VERTICAL MEMBERS AND STRAPPING (Sheet 1 of 2)

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NOTE

THIS PARTIAL LAYER, CONSISTING OF THREE STACKS, TWO ROWS, IS GIVEN AS A TYPICAL EXAMPLE OF AN UPPER PARTIAL LAYER RETAINED BY VERTICAL MEMBERS AND STRAPPING THESE PROCEDURES ARE APPLICABLE TO BOXCARS WITH EITHER METAL OR WOOD SIDE WALLS THE QUANTITY AND SIZE OF VERTICALS REQUIRED TO RETAIN ANY NUMBER OF STACKS AND ROWS ARE OBTAINED FROM THE TABULATION ON p 56 (WEIGHT RETAINED IN EACH ROW NOT TO EXCEED 8000 POUNDS)



DETAIL A
PARTIAL LAYER RETAINING FRAME
H = HEIGHT OF UNIT LOAD
W = DIMENSION OF UNIT LOAD CROSSWISE IN THE CAR

FIGURE 29 PARTIAL LAYER RETENTION WITH VERTICAL MEMBERS AND STRAPPING (Sheet 2 of 2)

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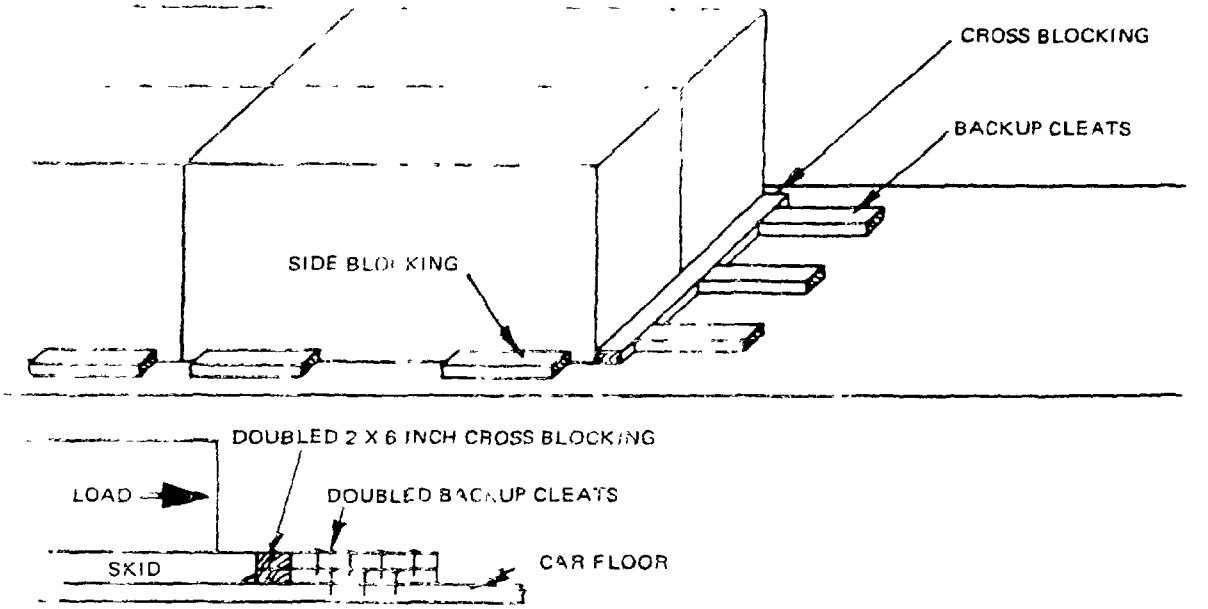


FIGURE 30 FLOOR BLOCKING

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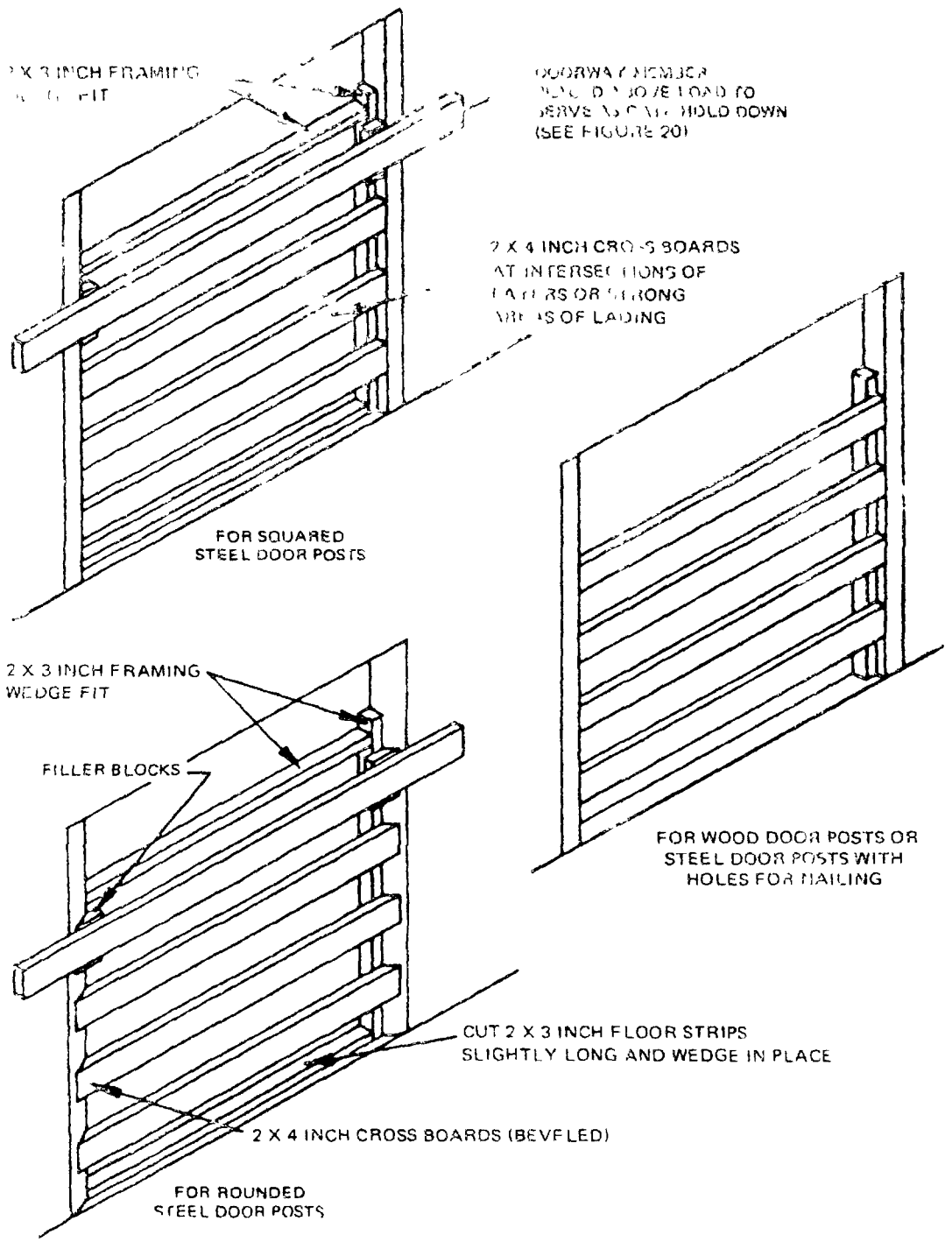


FIGURE 31. DOORWAY PROTECTION

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3.11.1 Doorway protection for cars equipped with plug-type doors. Each side of a plug-door boxcar may be equipped with a single plug door, double plug doors, or one plug door and one conventional door. Dunnage material must not be nailed to any plug door unless the door is provided with an adequate nailing strip, then dunnage may be nailed to the nailing strip when required. Stacks in the doorway that place a distributed load against the door must be unitized with two laterally applied 1-1/4-inch steel straps per stack, each tensioned and sealed with two double-crimped seals (see figure 32). The doorway area must be spanned with 2- X 6-inch members in all areas where the load results in concentrated forces against the door, thereby distributing these forces over the plug inner surface. If lumber of sufficient length to span plug doors is not available, random length material, doubled and spliced but with joints of splices offset, may be used.

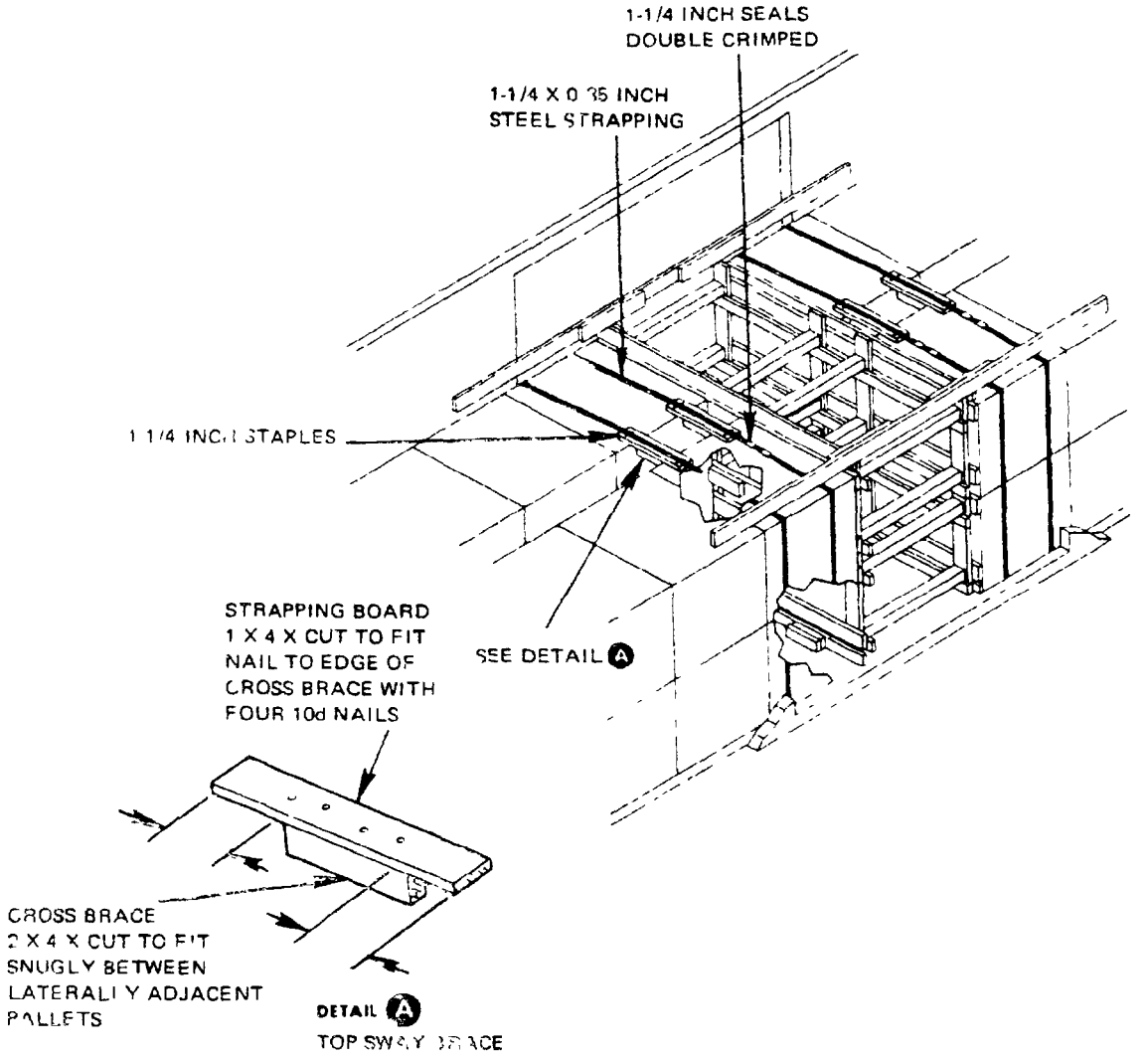


FIGURE 32. DOORWAY PROTECTION, STRAPPING METHOD

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5.10 Dunnaging design and procedure on flatcars

5.10.1 The loading, blocking, and strapping procedures described herein for flatcars are intended to provide general instructions for use when no MIL-STD dash number sheets exist. The basic difference between the loading of flatcars and the loading of boxcars is that on flatcars, all lengthwise, crosswise, and vertical forces must be restrained without the assistance of end walls or side walls. Because of this, the fundamental concept is to hold the load in position on the car platform with blocking and to hold the load down with strapping.

5.10.2 **Arrangement of load.** In planning the layout of the load on a flatcar, group the items being loaded into load groups, for example, four containers wide by three containers high. Determine the number of containers in a stack by considering the stability of the stacked items and the clearance required as shown on figure 7. Determine the number of containers across the width of the car from the width of the item and the width of the car available or to be ordered, allowing for the width of antichafing boards, side blocking, and at least a 3-inch clearance from the edge of the platform. Determine the number of load groups along the length of the car from the length of the item and the length of the car available or to be ordered, allowing for the length needed for intermediate blocking, end blocking, and about 6-inch clearance from the end of the platform. Consideration must also be given to the number of tiedown straps required and the stake pockets available. The total weight of the carload shall not exceed the load limit of the car, and the weight distribution of the load shall not exceed the limitations of figure 8, except with specific authority from the carrier.

5.10.2.1 **Load groups.** When more than one layer of containers is loaded, containers shall be strapped or bolted together to form a load group. Lengthwise and crosswise displacement shall be prevented with strapping, bolting, or appropriate blocking, tie rods, plates, or similar devices. Antiskid plates should be used between stacked wooden containers. The weight of a load group shall be no more than 35,000 pounds.

5.10.2.1.1 When there is more than one stack across the width of the car, the stacks should be strapped together so that the group of containers is an integral unit. Antichafing boards or frames should be placed between the stacks when required to prevent containers from rubbing against each other, thereby assuring a tightly bound load.

5.10.2.2 **Placement of load groups.** In planning the placement of load groups on a flatcar, the location of the side stake pockets on the car shall be taken into consideration since tiedown strapping must be fastened through stake pockets. Normally, stake pockets are positioned 42 inches on centers. The number of side stake pockets, therefore, depends

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upon the length of the car (see table VII). They may be inset or may extend beyond the edge of the platform. Position the load groups so that stake pockets will be available for the required tiedown strapping and so that there is sufficient space between the groups for the required intermediate blocking. A long container whose center of gravity is toward one end should be positioned so that the center of gravity is toward the lengthwise center of the car. Occasionally, car floor space may be saved and the car load increased by the use of pre-positioned blocking nailed to the car floor and located under the containers so that it butts against strong areas of the container and retains it from longitudinal and/or lateral movement.

Table VII

NUMBER OF STAKE POCKETS ON TYPICAL FLATCARS

Number of stake pockets on a side	Length of car
11	40 feet 6 inches to 41 feet 0 inch
12	41 feet 6 inches to 42 feet 6 inches
13	45 feet 0 inch to 46 feet 6 inches
14	47 feet 0 inch to 50 feet 6 inches
15	52 feet 0 inch to 53 feet 6 inches
17	60 feet 0 inch

5.10.2.3 **Ordering cars.** When ordering flatcars, stipulate the minimum platform length required and, when necessary, the minimum width, weight capacity, and number of stake pockets required. Do not order cars of greater length than required since this may increase the cost to the Navy. Only cars with sound floors shall be used. Cars with steel floor ends or exposed steel bolsters, or both, shall not be used.

5.10.3 **Blocking.** Blocking pieces are normally 2 x 6 inches in size and in 6-inch increments of length. They shall always be at least two layers thick, but three or more thicknesses may be used if additional height is needed to properly bear against the load. The length of the blocking pieces for intermediate blocking, side blocking, and end blocking depends upon the weight of the load. Each load group shall be adequately blocked in accordance with the requirement in this section, and the total amount of blocking shall be sufficient for the total carload weight.

5.10.3.1 **Nailing of blocking.** The strength of the blocking pieces is directly dependent upon the number of nails in each piece. One nail shall be used for every 6 inches of length or fraction thereof. For doubled blocking, use 30d nails in the first layer and 60d nails or

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spikes in the second layer. For side blocking, 30d nails may be used in both layers. When three or more layers of blocking are needed, use 30d nails in the first layer and 60d nails or spikes in each successive layer. For cross blocking, 8-inch spacing of nails is permitted.

5.10.3.2 **Cross blocking.** Cross blocking pieces, or headers, serve to distribute the load more evenly over the width of the car. While they obviously add strength to the blocking arrangement, their strength is not counted upon when determining the amount of end or intermediate blocking needed.

5.10.3.3 **End blocking or intermediate blocking.** End or intermediate blocking pieces are placed against cross blocking either at the ends of the last stacks or between the stacks. Align blocking with containers or container skids.

5.10.3.3.1 The number and length of end or intermediate blocking required depend upon the weight of the load group and the strength values given in table VIII. For a given number of blocking pieces, the required length is obtained by dividing the weight of the load group by the number of pieces being considered and selecting from the table a length whose strength is at least this amount. For blocking pieces of a given length, the number required is obtained by dividing the weight of the load group by the shear strength given in the table corresponding to the length being considered.

Table VIII

BLOCKING STRENGTH (2 X 6 DOUBLED)

Length (in.)	Nails per layer	Shear strength (lb)	
		End or intermediate blocking	Side blocking
18	3	3,500	2,500
24	4	4,500	4,000
30	5	6,000	6,000
36	6	7,000	9,000
42	7	8,500	13,500
48	8	10,000	17,500

5.10.3.3.2 Table VIII is based upon doubled end or intermediate blocking pieces nailed as described in 5.10.3.1. When blocking is tripled, add 6 inches to its length and one extra nail in each layer; when four layers high, add 18 inches to its length and three extra nails in each layer. At the ends of the flatcars, end blocking pieces less than 30 inches long should be avoided where possible.

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5.10.3.4 Side blocking. Side blocking pieces are placed against the skids or against the sides of the bottom container in a stack near its ends. They are usually doubled and positioned parallel to the length of the container and are not normally placed against the cross blocking pieces.

5.10.3.4.1 The number and length of side blocking pieces required depend upon the weight of the load group and the strength values given in table VIII. For a given number of side blocking pieces, the required length is obtained by dividing the weight of the load group by the number of pieces being considered and selecting from the table a length whose shear strength is at least this amount. For blocking of a given length the number required is obtained by dividing the weight of the load group by the shear strength given in table VIII corresponding to the length being considered.

5.10.3.4.2 Table VIII is based upon doubled side blocking pieces nailed as described in 5.10.3.1. When it is necessary for side blocking to act also as end or intermediate blocking add 6 inches to its length and one extra nail in each layer. When it is tripled, add 6 inches to its length and an extra nail in each layer. When side blocking is placed perpendicular to the length of the load, the lengths may be reduced 6 inches and one nail omitted.

5.10.4 **Strapping.** When a load group consists of several stacks of containers, strapping shall be used to band each stack, to tie together the top layers of containers of the stacks in each group, and to stay the load group to the flatcar, as shown in figure 33. Whenever possible, space the strapping evenly along the length of the containers. All seals shall be double crimped.

5.10.4.1 **Vertical strapping.** Each row of each stack must be banded with 1-1/4- x 0.035-inch steel straps. The number of straps required depends upon the weight of the containers being strapped together and the length of the containers. Use one strap for every 8,000 pounds, two straps for containers over 8 feet in length, and three straps for containers over 16 feet in length.

5.10.4.2 **Cross strapping.** The top layer of containers in each load group must be banded together with 1-1/4- x 0.035-inch steel straps. The number of straps required depends upon the weight of containers being strapped together and the length of the containers. Use one strap for every 8,000 pounds and at least two straps for containers over 16 feet in length.

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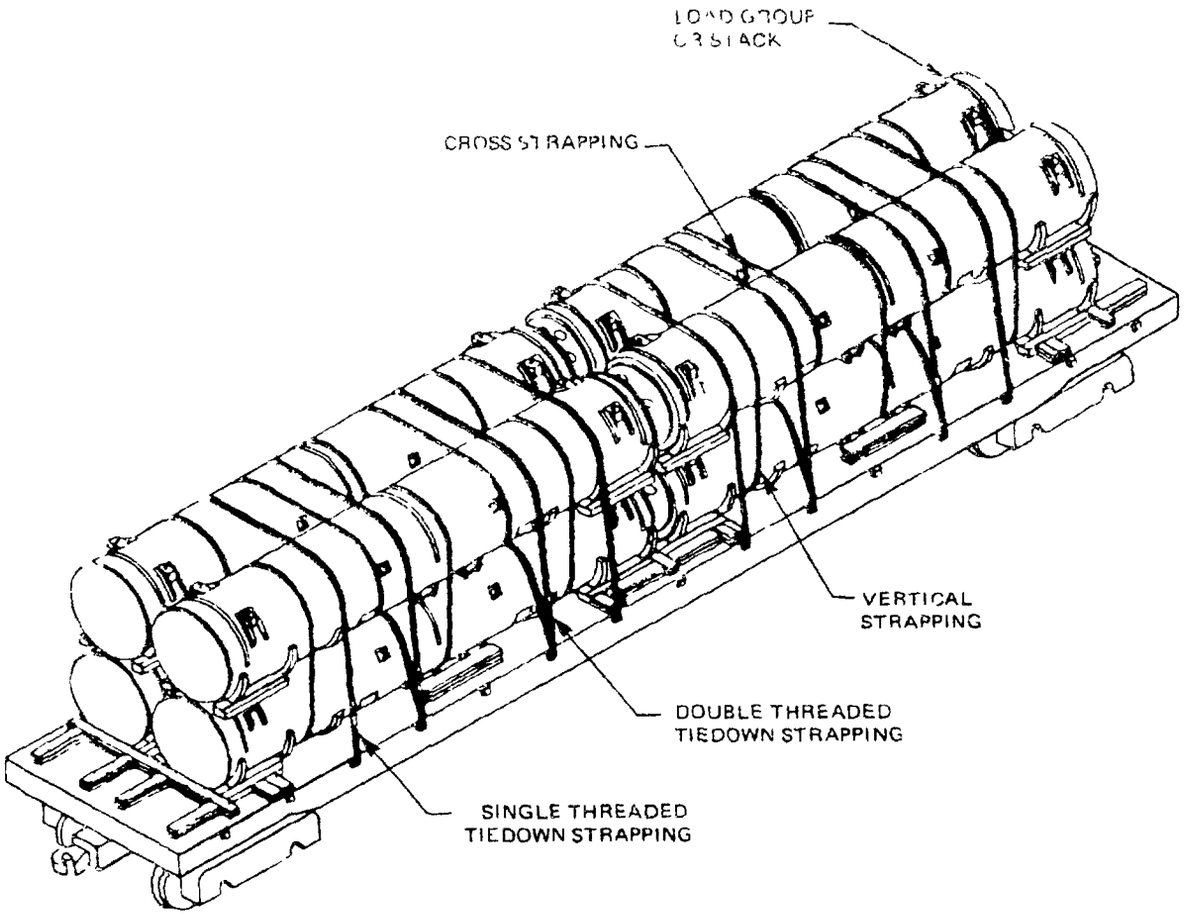


FIGURE 33 TYPICAL METHOD OF STRAPPING CONTAINERS TO A FLATCAR

5 10 4 3 **Tiedown strapping** Load groups must be securely stayed to the platform of the flatcar using 2- x 0 050-inch steel straps marked with the letters "AAR" and with the manufacturer or distributors name abbreviated name, registered trade mark or symbol All tiedown strapping must have two seals and each seal must be double crimped Tiedown straps should be anchored to stake pockets, wherever possible When this is not possible anchor points must be of sufficient width to receive a 2-inch strap Do not use swivel ring-type anchor devices Stake pockets shall be padded with a stake pocket pad made of 2- x 0 050-inch strap positioned under and sealed to the tiedown strap with one seal Alternatively two commercial stake pocket protectors under each stake pocket may be used Edge protectors shall be used under tiedown strapping at the top edges of containers or [illegible] with sharp edges

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5 10 4 3 1 The number of straps required depends upon the weight of the load group and a factor given in table IX This factor is related to the length of the container and the type of shock mounting of the item within the container, as follows

(a) Hard item mounting includes bolting and similar rigid mounting of the item within the container

(b) Semihard item mounting includes hard rubber mounts, shear mounts, and similar devices which permit limited and damped movements of the item within the container

(c) Soft item mounting includes springs, cushioning material, and other devices and materials which permit significant and repeated movements of the item within the container

5 10 4 3.2 To determine the minimum number of tiedown straps required divide the weight of the load group by 8 000 and then multiply by the appropriate factor in table IX (For fractional results use the next highest number of straps)

Table IX

FACTOR FOR DETERMINING
NUMBER OF TIEDOWN STRAPS

Type of item mounting (see 5 10 4 3 1)	Container length	
	Under 10 ft	Over 10 ft
Hard	2 0	1 2
Semihard	2 4	1 6
Soft	3 0	2 2

5 10 4 3 3 When there are an insufficient number of anchor points for the number of tiedown straps required, the strap should be double threaded over the load as follows thread the strap up through stake pocket on one side of car, over the top of the next-to-last layer of containers, down to and through stake pocket on opposite side of car, and back over the top of the load where it is tensioned and doubled sealed

5 10 5 **Placards** For explosive loads placards must be placed on placard boards centrally located on both ends and both sides of car If the car is not so equipped make 16- x 24-inch placard boards nailed to two 2- x 4 x 24-inch uprights which are to be nailed to dunnage with four 10d nails in each upright

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5.11 MIL-STD dash number sheets

5.11.1 The MIL-STD-1325 dash number sheets (i.e. MIL-STD-1325-1, 2, 3 etc.) are a series of detailed instructions for the loading, blocking and bracing of specific ammunition explosives or associated items. As they are published, the MIL-STD-1325 dash number sheets supersede the WR-52 slash number sheets now in use. Until the superseding MIL-STD-1325 dash number sheet is published, the WR-52 slash number sheet forms a part of this standard.

5.11.2 Identification numbering of these MIL-STD sheets consists of the basic MIL-STD-1325 designator followed by a dash number for each group of sheets, or in case of WR sheets the numbering identification will be the WR-52 designator followed by a slash number for each group of sheets. These numbers are in numerical sequence without duplication.

5.11.3 The Naval Weapons Handling Laboratory (NWHL) at the Naval Weapons Station Earle, Colts Neck, N.J., develops and maintains MIL-STD dash number sheets. MIL-HDBK-236 provides an alpha-numerical index of items covered by the MIL-STD dash number sheets and the WR slash number sheets. This index is updated quarterly. Recommended corrections, additions or deletions should be addressed to Commanding Officer, Naval Weapons Station Earle (NWHL), Colts Neck, N.J. 07722.

Custodian
Navy – OS

Review activities
Navy – AS, OS, MC

User activities
Navy – SH, SA

Preparing activity
Navy – OS
(Project No. 8140-N132)

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Appendix

10 QUALITY ASSURANCE PROVISIONS

10.1 **Scope** This appendix covers impact testing, trial shipment, and inspection of railcar loads of ammunition, explosives, and other dangerous articles (AEDA)

10.2 **Purpose** This appendix is intended to establish standard procedures for the following

(a) Railcar impact tests of carloads or less than carloads of unique items of lading and new methods of dunnaging

(b) Trial loadings and trial shipments of carloads or less than carloads of lading new methods of dunnaging, or shipments that are presenting particular difficulties

(c) Inspection of carloads and less than carloads that have an approved MIL-STD dash number sheet

(d) Inspection of carloads and less than carloads that do not have an approved MIL-STD dash number sheet

(e) Inspection of mixed carloads and less than carloads

10.3 **Application** When specified, the material contained in this appendix is a mandatory part of this standard

10.4 **Responsibility for railcar impact tests, trial shipments, and inspections**

(a) The performance of railcar impact tests is the responsibility of NAVSEASYSCOM and NWS Earle, Naval Weapons Handling Laboratory (NWHL)

(b) The performance of trial loadings and trial shipments is the responsibility of NAVSEASYSCOM NWS Earle (NWHL), the shipping activity and the receiving activity

(c) Inspection of all carloads and less than carloads is the responsibility of the shipping activity

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10.5 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) **First article inspection.** First article inspection consists of examinations and tests conducted, prior to general use, on proposed loads (inert or prototype), to ensure that the design is such that the load is capable of withstanding the rough handling test requirements of this standard (see 10.6).

(b) **Quality conformance inspection.** Quality conformance inspection consists of those examinations accomplished on approved loads prior to shipment, to ensure that the lading is loaded in accordance with the approved carloading plan and the methods specified in this standard (see 10.11).

10.6 **First article inspection.** First article inspection shall consist of the tests specified in 10.7 and 10.8 and the examinations specified in table X. First article inspection of carloads or less than carloads will not be required if the lading and loading procedures are in substantial conformity with the existing rules and regulations.

10.6.1 **First article sample.** The first article sample shall consist of one prototype load or a load of inert material representative of that to be shipped, placed on a railcar exactly as shown on the proposed loading plan.

10.6.2 **Use of dummy loads.** Dummy loads may be used during the development program when inert loaded end products are not available. The dummy shall have the following characteristics identical to those of the objects being simulated:

(a) Envelope dimensions.
(b) Weight, center of gravity, and radii of gyration in the three principal axes.

10.7 **Railcar impact test.** When, in the opinion of NAVSEASYSCOM, NWS Earle (NWHL) and the Bureau of Explosives, a proposed carload is considered unique because of the characteristics of the load, structural features of the lading, or the carloading procedures, a railcar impact test shall be conducted to prove the safety and adequacy of the proposed carloading plan. Impact tests shall be coordinated with Military Traffic Management and Terminal Service, NAVSEASYSCOM, Bureau of Explosives, and NWS Earle (NWHL).

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Table X

VISUAL EXAMINATION OF RAILCAR LOADS

Examination	Applicable paragraph	Acceptance criteria
Boxcar		
Type, size and capacity	5.2.1 5.2.2	The type, size and capacity of the railcar is suitable for the shipment to be loaded
Cleanliness	5.5.2	The interior of the car is clean
Protrusions and debris	5.5.2	There are no protruding nails, screws or bolts and all debris and dunnage have been removed
Serviceability of cargo spaces	5.3.1	Floor, walls and roof are tight and free of holes, cracks, loose boards, and decayed spots
Fire proofness	5.3.1	Roof, walls and floors are free of any combustible liquid or solid
Metal floors	5.3.2	Cars with metal floors are not being used for shipping bulk explosives that are liable to leakage of dust, powder or vapor
Vehicle loading device	5.3.2	The railcar is not equipped with a vehicle loading device
Pin type devices	5.3.2	Pin-type devices on railcar equipment are free from obstructions and are in good working order
Equipment	5.3.2	The full complement of DODX railcar equipment is present
Rack-type DODX cars	5.3.2	Rack-type DODX railcars are not being used for over-the-road shipments (outside of the activity)
Doors	5.2.2.2	End door railcars are not being used for transporting hazardous materials. Doors are of sufficient size for largest item
Tightness of doors	5.3.1	Doors fit tightly when closed and are capable of being locked and sealed
Flatcar		
Size and capacity	5.10.2.3	The size and capacity are the minimum required consistent with intended use
Flooring	5.10.2.3	Flooring is sound with no missing boards
Steel floor ends or exposed steel bolsters	5.10.2.3	The flatcar does not have steel floor ends or exposed steel bolsters

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Table X (contd)

Examination	Applicable paragraph	Acceptance criteria
Mechanical Condition		
Boxcars and Flatcars	5 3 3	All running gear of the car is in good mechanical condition
Compatibility of Mixed Loads		
Mixture of explosives	5 2 1 1	The shipment does not contain any combination of explosives or other hazardous materials which are prohibited by DOT regulations from being loaded transported or stored together
Other hazardous articles	5 2 1 1 4 2 3 1	In addition to incompatible explosives, the following articles are not being shipped in DODX utility loader railcars unless permission has been obtained from Commander, Naval Sea Systems Command (SEA-044, SEA-045, SEA-65161) flammable liquids flammable solids and oxidizing materials acids and other corrosive liquids radioactive materials, and any other material that is likely to contaminate impair or otherwise damage the car so as to prevent it from being certified for use with Class A explosives
Hazard Classification of Load	4 2 3 1	The hazard classification of the load has been determined prior to loading and is reflected on the loading plan
Dunnage Materials		
Lumber	5 7 3	Lumber conforms to the requirements of this standard
Spikes	5 7 4	Spikes are of the round wire type
Nails	5 7 4	Nails are installed in accordance with the loading plan or section 5 of this standard
Strapping	5 7 5	Strapping conforms to the requirements of this standard and is installed as specified on the loading plan or section 5 of this standard
Mechanical dunnage (DODX cars)	5 8	Mechanical dunnage is used as required All parts are available and in proper operating condition Maximum allowable loads are not exceeded

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(contd)

Examination	Applicable paragraph	Acceptance criteria
Boxcar load		
End gates	5 9 2	End gates are used in cars with bowed or unlined metal ends
		End gates are used when distributing concentrated loads over the end of a railcar for such items as uncreated bombs or empty projectiles
		End gates are installed in accordance with this standard
Sway bracing	5 9 3	Sway bracing is installed when lading does not completely fill the car crosswise
Floor sleepers	5 9 3 1	Sleepers run parallel to the long dimensions of the railcar and are of at least 2- x 4-inch lumber
Longitudinal gating	5 9 3 2	Construction of gating is similar to that of strut-type center gates except that it is placed lengthwise in the railcar, either along the centerline or along the railcar wall
Top-of-the-load sway bracing	5 9 3 3	The bracing is nailed or strapped to cross members which are used to brace the tops of the containers
		For heavy wooden boxes or crates the bracing is nailed to support pieces which are nailed to the boxes or crates
Interpallet sway bracing	5 9 3 4	The first layer is braced with longitudinal sleepers nailed to the floor against the pallet
		For the second and additional layers, spacer frames are placed between posts of pallets across the railcar
		Spacer frames are inserted under pallet decks and rest on pallet runners or are nailed to spacer frame supports
		The width of the spacer frame is 1 inch less than the distance between pallet posts across the railcar
Holddown bracing	5 9 4	Holddown bracing is installed when required in accordance with loading plan

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Table X (contd)

[illegible]nation	Applicable paragraph	Acceptance criteria
Separation gates	5 9 5	Divisional gates are installed where required in accordance with loading plan
Center gates	5 9 6	Wedge-type center gates are used when the space at the center of the railcar is under 20 inches and wedges can be driven (height approx 48 inches) Strut-type center gates are used when the space at the center of the railcar is 20 inches or greater When struts exceed 48 inches in length horizontal and vertical tie bars are added Center gates are not nailed to car floors walls holddowns or doorway protection but are left free to move with the load
Vertical members	5 9 6 1	At least four vertical members are used and are located in line with solid surfaces of the unit loads, containers, packages, or items comprising the car load The outside vertical members of wedge-type gates are at least 2 inches in from the ends of horizontal members to allow space for the tie cleats Each vertical member is nailed to a horizontal member with three 10-penny nails
Horizontal members	5 9 6 2	The length of horizontal members of the center gate are 1 inch less than the inside width of the car Horizontal members are located at or near the top and bottom of the load and are in line with solid surfaces of the load The bottom horizontal members are located 4 inches above the floor when used with palletized unit loads
Strut cleats	5 9 6 3	The length of strut cleats may be 1 inch less than the inside width of the railcar but are long enough to extend past the outside verticals Cleats are positioned so that the bottom of the strut will be about 1 inch above the bottom of the horizontal gate member Strut cleats are not used on wedge type gates

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Table [illegible] (Contd)

Examination	Applicable paragraph	Acceptance criteria
		Strut cleats are nailed at the joint of the vertical member with three 10-penny nails
Struts	5 9 5	Struts are located at or near top and bottom of gate and at or near ends of gate and are aligned with horizontal and vertical members wherever possible
		Intermediate struts are spaced as equally as alignment permits
		Sufficient struts are used to distribute the load evenly
		Struts normally are fabricated from two pieces of 2 × 6-inch lumber When such struts are used the two 2- × 6-inch members are nailed together with one 10-penny nail every 5 inches Single 4- × 4-inch struts may be used in place of double 2 × 6-inch members
		Single 2- × 6-inch struts are used for light loads
		Struts are toenailed to gate vertical members with two 12-penny nails
		Struts are not nailed to floors or walls
Tie bars	5 9 6 5	Horizontal and vertical tie bars are nailed at the midpoints of the struts when the length of the struts exceed 48 inches
		Tie bars are nailed to struts with three 16-penny nails at each joint
Gate holddowns	5 9 6 6	The gate holddowns are nailed across the doorway to prevent the center gate from riding upward and bucking
		Holddowns are constructed of 2- × 6-inch lumber the length of which is 24 to 48 inches wider than the doorway
		Holddowns are nailed to railcar walls using five 10-penny nails at each end
End bracing	5 9 7	End bracing is used for less than carload shipments
		End bracing is not used in lieu of a center gate structure when the size of shipment will permit use of center gates

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Table X (contd)

Examination	Applicable paragraph	Acceptance criteria
		end bracing does not exceed the weights specified for the various types of bracing
		The angle which diagonal braces make with the floor do not exceed 45° and are not less than 30
Diagonal over bracing	5.9.8	Bracing is installed when required
		Diagonal members are positioned so that the angle between the wall and the diagonal does not exceed 45° nor is less than 30°
		The ends of the diagonals are double beveled for good bearing against bracing members and cleats
		Car lining is reinforced by pieces of lumber nailed to the side of the railcar
		Nails are driven into railcar studding insofar as practicable
Floor blocking	5.9.9	Floor blocking is placed tightly against the load and securely nailed or bolted to the car floor
		Floor blocks are at least nominal 2 x 4 inches
		The floor block is at least as thick as the skid member where blocking butts against a container or loaded skid
		Crosswise floor blocks extend the full width of the load and are provided with backup cleats of the same nominal thickness
		Backup cleats are placed in line with skid members
		Backup cleats are extended over three or more floor boards to ensure proper nailing
	5.9.9	Length of backup cleats are extended when cars with nailable steel floors are used
Doorway protection	5.9.10	Methods of applying doorway protection in railcars with steel-jacketed door-posts and wooden doorposts are in agreement with this standard
		Cross boards are sufficiently close to the floor and so constructed as to prevent units from falling or rolling out of railcar during transit

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Table VI (cont'd)

Examination	Applicable paragraph	Acceptance criteria
Plug doors	5.9.11	Dunnage material is not nailed to any plug door except where an adequate nailing strip is provided. Dunnage may be nailed to the strip when required. Where load against plug doors is distributed, bulkheads in doorway area are unutilized with steel strapping. Where load against plug doors is concentrated, doorway area is spanned with 2" x 6"-inch members, all areas with concentrated forces against the door.
Flatcar Load		
Load groups	5.10.2.1 5.10.2.2	If more than one layer of containers is loaded, the containers are strapped or bolted together as an integral unit. Lengthwise and crosswise displacement is prevented by the use of strapping, bolting, appropriate blocking, tie rods, plates, or similar holding devices. Antiskid plates are used between wood containers or crates. The weight of the load group does not exceed 35,000 pounds. If there is more than one row across the width of the railcar, rows are strapped together so that the group of containers acts as an integral unit. Antichafing boards or frames are placed between rows to prevent containers from rubbing against each other.
Blocking	5.10.3	Blocking pieces consist of at least two layers of 2" x 6"-inch lumber nailed together. The length of the blocking pieces is adequate for the weight of the load in accordance with table VIII.
Nailing of blocking	5.10.3.1	One nail is used for every 6 inches of length or fraction thereof. For double blocking, 30-penny nails are used in the first layer and 60-penny nails in the second layer. However, for side blocking, 30-penny nails are used in both layers.

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Table (contd)

Examination	Applicable paragraph	Acceptance criteria
		When three or more layers of blocking are used 30-penny nails are used in the first layer and 60-penny nails or spikes are used in each successive layer
Cross blocking	5 10 3 2 5 10 3 3	End blocking or intermediate blocking is placed against the cross blocking between load groups or at the ends of the last group
		End blocking or intermediate blocking is aligned with containers or container skids
		The number and length of end blocks and intermediate blocks is in accordance with table VIII
Side blocking	5 10 3 4	The number and length of side blocking pieces is in accordance with table VIII
		Side blocking pieces are placed against skids or against the sides of the bottom container in a stack
		Side blocking is normally doubled and positioned parallel to the length of the container and not normally placed against crossblocking pieces
Strapping	5 10 4	When a load group consists of several rows of containers, strapping is used to band each row to tie the top layer of containers together in each group and to strap each load group on the flatcar
		All strap seals are double crimped
Vertical strapping	5 10 4 1	Each stack is banded with 1-1/4- x 0 035-inch steel straps
		One strap is used for every 8 000 pounds of lading
		Two straps are used for containers over 8 feet long
		Three straps are used for containers over 16 feet long
Cross strapping	5 10 4 2	The top layers of containers in each load group are banded together with 1-1/4- x 0 035-inch steel straps
		One strap is used for every 8 000 pounds of lading
		Two straps are used for containers over 16 feet long
Tiedown strapping	5 10 4 3	Load groups are securely staved to the platform of the flatcar using 2 x 0 050-inch steel straps properly marked

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Table IX (contd)

Examination	Applicable paragraph	Acceptance criteria
		The number of tiedown straps is in accordance with table IX data All tiedown strapping has two seals at each strap junction, and each seal is double crimped Swivel-type rings are not used as anchor devices for strapping Tiedown straps are anchored to stake pockets wherever possible Stake pockets are padded with a 2 x 0.50-inch strap positioned under and sealed to the tiedown strap with one seal Edge protectors are used under tiedown strapping at the top edges of wooden containers or similar items with sharp edges
Prior to Release of Loaded Railcar	5.6.1	
Storage of equipment	(h)	DODX car equipment is stored in accordance with requirements
Shipping of documents	(j)	Shipping documents are attached to the center gate end bracing or some other conspicuous place before the railcar doors are closed and sealed
Seals	(l)	Car is sealed with Navy strap-type seal
Placards	(m)	Each railcar used to transport hazardous materials is appropriately placarded by the shipping activity in accordance with DOT regulations. Placards shall be securely attached to each end and to each side of the railcar or to placard boards provided for the purpose. Placard requirements for railcars are specified in R. M. Graziano's Tariff.

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10.7.1 **Railcar impact test requirements** On a reasonably straight track free of switches set up a minimum of 5 buffer cars with slack out of couplers and all brakes set. Total weight of the buffer string shall be at least 250 short tons. The specimen cars, loaded in accordance with the proposed method (inert loaded ordnance may be used) shall roll freely into the buffer string successively at speeds of 4, 6, and 8 miles per hour (mph). The specimen car shall then be reversed and impacted at 8 mph. Suitable instrumentation shall be employed to measure impact speeds and to record the before and after condition of lading.

10.7.2 Upon completion of tests there shall be no damage to the contents, toppling, nor movement of the load likely to produce damage to the contents or to railroad equipment.

10.8 **Trial shipments** A trial shipment is conducted to verify that loading instructions do, in fact, provide the protection required. Trial shipments shall be coordinated with the originating carrier and in the case of ammunition explosives and dangerous articles with the Bureau of Explosives and with NAVSEASYSCOM (SEA-044, SEA-045, and SEA-65161).

10.8.1 **Trial shipment procedure** The following steps shall be followed by the organization responsible for the conduct of the trial.

(a) Load the packaging system on the railcar exactly as required by the carload plan.

(b) Record the position of the packaging system together with the dunnage and fastenings used to constrain it. (Sketches and photographs, or both).

(c) Arrange for the transport of the packaging system over a suitable prescribed route. The car containing the packaging system shall be interchanged between carriers made up into at least three different trains.

(d) Upon receipt of shipment and prior to each loading or unloading, inspect the packaging system and the constraining dunnage and fastenings. Record any evidence of damage or inadequacies.

(e) After unloading, the contained item shall be tested to ascertain any change in its original operating or functional characteristics. Any indication of shipping damage shall be recorded accordingly.

10.8.2 **Test acceptance criteria** There shall be no change in the original physical and functional characteristics of the contained components or assemblies. There shall be no indication of any transmission of excessive vibration or shock stresses to any component or assembly. The packaging system shall not display any evidence of functional degradation. Negligible material damage to the packaging system, such as chipped paint, minute dents, or scratches, shall not be deemed to be sufficient cause of rejection.

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10.9 Test report. A report shall be prepared as a separate document or as a part of the request for approval. This report shall detail all tests performed and give complete results of the tests, including any minor damage which may not be considered as cause for rejection. Photographs of the unit load before and after testing shall be made a part of this report. Additional photographs showing any special test setups shall also be included in the report.

10.10 **Classification.** A new explosive, except samples for laboratory examination by the Bureau of Explosives, may not be shipped unless it has been classified as specified in 4.3.3.1.

10.11 **Quality conformance inspection.** Quality conformance inspection shall consist of visual examinations on railcars and loads to be accomplished before, during, and after loading to ensure that the railcar is safe for transportation of the intended load, the loading procedures are in accordance with approved standards, and the materials are loaded and secured in accordance with approved loading plans and practices.

10.12 **Examination of carloads.** Carloads shall be examined as follows:

(a) Carloads and less than carloads that have an approved MIL-STD-1325 dash number sheet shall be examined to assure that the loading has been accomplished in accordance with the approved document. Particular emphasis shall be placed on assuring that the lading, when called for, is tightly positioned against the end walls and side walls of the car and that dunnage tightly fills all void spaces longitudinally and laterally.

(b) Carloads and less than carloads that do not have an approved MIL-STD-1325 dash number sheet and all mixed carloads and less than carloads shall be examined to assure that the loading has been accomplished in accordance with the applicable paragraphs of this standard as referenced in table X and the rules and regulations of the Department of Transportation, the Bureau of Explosives, and the Association of American Railroads.

10.13 **Rejection criteria.** Nonconformance with the acceptance criteria listed in table X for corresponding examination shall be cause for rejection of the railcar or carload as applicable. Minor repairs are permitted in order to bring the railcar to an acceptable level of serviceability.

10.14 **Railcar signoff.** Prior to releasing the railcar to the carrier, the load inspector shall complete items 14 through 30 of NAVORD Form 8023 and sign the form. If the railcar has been loaded by the shipping activity, part 2 of the railroad car certificate must be signed by the representative of the shipping activity and by the carrier's representative. If the railcar has been loaded by the carrier, only the carrier's representative shall sign part 2 of the certificate. All deficiencies shall be corrected before the railcar is released to the carrier for shipment to the destination.

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STANDARDIZATION [illegible] IMPROVEMENT PROPOSAL

OMB Approval No. 22-R-[illegible]

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DOCUMENT IDENTIFIER AND TITLE

MIL-STD-1325A RAILCAR LOADING OF HAZARDOUS MATERIALS

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   a. GIVE PARAGRAPH NUMBER AND WORDING.

b. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO (If "Yes" in what way?)

4. REMARKS

SUBMITTED BY (Printed or typed name and address - Optional)	TELEPHONE NO DATE

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